

When label-free morphology is sufficient for targeted cellular measurements

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Abstract

Label-free morphology captures cellular responses to perturbation, but can it preserve targeted readouts of molecular, organellar and functional states? We assess measurement sufficiency using nearly 10 million paired phase-reporter observations from an A549 optical perturbation atlas spanning 1,000 gene knockouts, 52 fluorescent reporters and 73 screens. Ten prediction methods differed in their recovery of reporter phenotypes across held-out imaging fields, genes and screens. Pairing controls identified information about individual-cell reporter states beyond average perturbation responses and measured covariates. Perturbation ranking often persisted despite a narrower spread of predicted response magnitudes: 11 reporters retained less than half the observed magnitude variance. Replacing measured responses to held-out gene knockouts with predictions preserved a median of 79% of the strongest perturbation hits, while retention of leading functional terms varied across reporters. Atlas-derived response criteria identified 30 quantitative and 7 ranking proxies for a fixed prediction ensemble. Reproducibility of the reference measurements and screening context further constrained interpretation. In an independent primary-hepatocyte resource, brightfield predictions selected 10 of the 11 compounds with the largest measured metabolic-activity losses among 217 held-out compounds, supporting prioritization for follow-up measurement. Measurement sufficiency therefore depends on the target, predictor, experimental context and scientific task, rather than predictive correlation alone.

1

2 Introduction

3 AI-powered virtual-cell models aim to connect multimodal observations of cellular state with predic-
4 tions of biological responses¹. Their use as virtual instruments raises a practical biological question:
5 when can a predicted readout support the experiment for which a direct measurement would otherwise

6 be needed? Functional-genomics screens link genetic perturbations to changes in cell state^{2,3}. High-
7 content microscopy measures molecular abundance, organelle organization and cell morphology^{4,5}.
8 Perturb-seq measures transcriptional responses^{6,7}, whereas Perturb-CITE-seq also measures changes in
9 cell-surface proteins⁸. Yet individual experiments observe only a subset of these states, constrained by
10 optical channels, reagents and throughput. Computational inference could extend this coverage if the
11 predicted readouts retain the biological information required by the experiment.

12 Label-free morphology provides a concrete setting for this question. Phase images can be collected
13 repeatedly without target-specific labels⁹, while pooled optical screening links cellular images to pertur-
14 bation identity¹⁰. Recent platforms extend image-based genetic screening to large-scale morphological
15 profiling^{11,12} and primary cells and tissues¹³. Quantitative image analysis converts cellular appear-
16 ance into profiles of morphology and subcellular organization^{14,15}. These profiles can reveal similarities
17 among perturbations¹⁶ and predict measures of cell health¹⁷.

18 Previous work has inferred fluorescent labels from unlabelled images^{18,19}, including channel predic-
19 tion for high-content screening^{20,21}. Related methods connect cellular images with molecular profiles^{22,23}
20 or predict cell-surface protein abundance from RNA measurements²⁴. Comparisons of morphological
21 and transcriptional perturbation profiles also reveal shared and complementary structure^{25,26}. These ca-
22 pabilities motivate targeted prediction, but they address different properties: a reporter can distinguish
23 biologically meaningful perturbations, share structure with morphology, or be recoverable from mor-
24 phology for an unseen perturbation. Establishing a computational substitute further requires knowing
25 which uses of that reporter are preserved.

26 We use *measurement sufficiency* to ask whether replacing a measured phenotype preserves the evi-
27 dence required for a specified scientific task, following the principle of application-appropriate valida-
28 tion in quantitative bioimage analysis²⁷. Prioritizing strong perturbations requires reliable ordering and
29 hit recovery; comparing effect sizes additionally requires quantitative fidelity; interpreting functional
30 programmes requires retention of the relevant downstream analysis. A predictor can perform well on
31 one task and poorly on another. Agreement can also reflect perturbation-group averages or coarse co-
32 variates such as cell count²⁸, while a relation learned in one cellular context can change in another²⁹. The
33 reference response must itself be reproducible enough to evaluate substitution. Predictive correlation is
34 therefore one component of a measurement assessment, alongside response fidelity, target reliability
35 and preservation of the intended biological analysis. Distinguishing these components clarifies what an
36 inferred measurement can support and where additional measurement is needed.

37 We investigate these requirements in a public A549 optical perturbation atlas containing 1,000 gene
38 knockouts, 52 fluorescent reporters and 73 screens³⁰. The reporters span ten biological systems, in-
39 cluding readouts of subcellular organization, signalling and cellular stress. Quality-controlled linkage
40 provides nearly 10 million paired phase-reporter observations, allowing fluorescence-derived pheno-
41 types to be predicted from phase morphology measured in the same cell after perturbation. Reporter
42 measurements from training perturbations provide the paired reference data. Same-cell pairing con-
43 trols test whether predictions capture individual-cell states beyond average perturbation responses and
44 measured covariates, while repeated screens test transfer to another acquisition context. We compare
45 ten prediction methods, then distinguish response ranking from quantitative fidelity and examine the
46 reproducibility of the reference measurements. Replacing responses to held-out gene knockouts with
47 predictions directly tests retention of strong perturbation hits and functional analyses. Independent ap-
48 plications extend the assessment to other measurement designs, including brightfield-guided prioritiza-
49 tion of metabolic-activity loss in primary hepatocytes. Together, these analyses identify which biological
50 uses of each readout are supported by prediction and which require further measurement.

51 Results

52 The atlas defines 99 related but non-interchangeable measurement tasks

53 The source atlas links pooled genetic perturbations to phase and targeted-fluorescence phenotypes in
54 the same cells (Fig. 1a,b). Quality-controlled linkage yielded 7,344,374 distinct phase cells and 9,996,286
55 phase-reporter observations; one phase cell could be linked to more than one reporter measurement.
56 Each cell was described by 172 engineered phase features. The paired fluorescence images provided 20–
57 72 phenotype measurements, or endpoints, per reporter, describing signal intensity and spatial distribu-

tion. Together, the 52 reporters contributed 1,604 reporter-specific endpoints rather than one common measurement panel.

The acquisition graph connected 52 reporters from ten biological systems to 73 screens through 99 observed reporter–screen assays (Fig. 1c). Each screen contained one to seven reporters; unacquired combinations were missing assays, not negative observations. Only 34 reporters supported a destination–screen test. Unlike collections that repeat a common panel^{11,31}, this sparse topology requires separate reporting by target, screen and acquired assay: a screen summary reflects both its reporter composition and acquisition context. Reporter programmes and exact-linked FeRhoNox, LAMP1 and pRb examples illustrate why the assays are not interchangeable replicates (Fig. 1d,e).

Predictor choice affects recovery of targeted phenotypes across evaluation settings

We compared ten predictors to determine how method choice affects the recovery of targeted phenotypes: Ridge regression, gradient-boosted trees, CatBoost³², a reporter-specific multilayer perceptron (MLP), TabM³³, scButterfly³⁴, a shared residual MLP, MultiTab, MIDAS³⁵ and scPair³⁶. These methods include models fitted to one reporter at a time, models trained jointly across reporters and methods developed to predict one measurement modality from another. All used the same phase representation, observed target endpoints and predefined partitions, with method-specific training scopes and adaptations documented in Methods.

The evaluation separated three scientific questions: prediction in an observed perturbation context, reconstruction of responses to unseen knockout genes, and transfer to a new acquisition environment. We tested these by holding out fields, genes or an entire destination screen, respectively (Fig. 2a). Within each test, cell-level correlation measured exact-linked phenotype recovery, whereas knockout–response correlation measured reconstruction of control-relative perturbation effects. Reporter, screen and reporter–screen summaries then showed where performance was retained. This distinction matters because a method ranking is conditional on what is predicted and how generalization is tested^{37,38}.

Method choice produced statistically and practically distinct recovery in the paired reporter comparisons. Eleven of 54 comparisons against the prespecified MLP met both the Holm-adjusted $P < 0.05$ criterion and an absolute median paired difference of at least 0.02. All eleven favoured the MLP over Ridge, scButterfly or MIDAS in field or gene evaluations (Fig. 2b). For held-out-gene knockout responses, median MLP recovery was 0.768, compared with 0.691 for scButterfly and 0.614 for MIDAS; the corresponding median within-reporter gains were 0.081 and 0.133 ($n = 52$; both Holm-adjusted $P < 2 \times 10^{-8}$). Ridge had a median recovery of 0.736; the paired MLP advantage was statistically significant but below the 0.02 effect criterion (median difference 0.018; Holm-adjusted $P < 2 \times 10^{-8}$). Comparisons not meeting both criteria do not establish method equivalence.

Descriptive rankings also depended on the test and reporting unit. The MLP had the highest reporter-level knockout-response median with held-out fields (0.715), whereas Ridge had the highest whole-screen median (0.730). The latter exceeded the MLP median of 0.677, but the median paired Ridge-minus-MLP difference was -0.002 across the same 34 reporters (Holm-adjusted $P = 1$), illustrating why pooled medians cannot substitute for target-matched inference. Screen and assay views localized performance across the acquisition topology (Fig. 2c,d); field and gene cell-level views reweighted reporter scores rather than providing independent assay-level evaluations.

The recoverability landscape was summarized by the unweighted median of the ten method-level scores. For response fidelity and counterfactual replacement, we instead evaluated a fixed prediction ensemble: the element-wise median of aligned out-of-fold profiles. Measurement decisions and their relation to retained utility used recoverability calculated from these same ensemble predictions. This distinguishes a summary of method performance from assessment of a particular inferred readout, without selecting a predictor separately for each target.

Target value, cross-modal overlap and directed recoverability are distinct properties

Median knockout-response scores across the ten methods varied widely across reporters and screens (Fig. 3a). For descriptive summaries we divided scores into low ($r < 0.60$), intermediate ($0.60 \leq r < 0.75$) and high ($r \geq 0.75$) recoverability strata. In the held-out-gene test, 8 of 52 reporters, 5 of 73 screens and 12 of 99 assays had low recoverability. Under whole-screen transfer, low recoverability occurred in 12 of

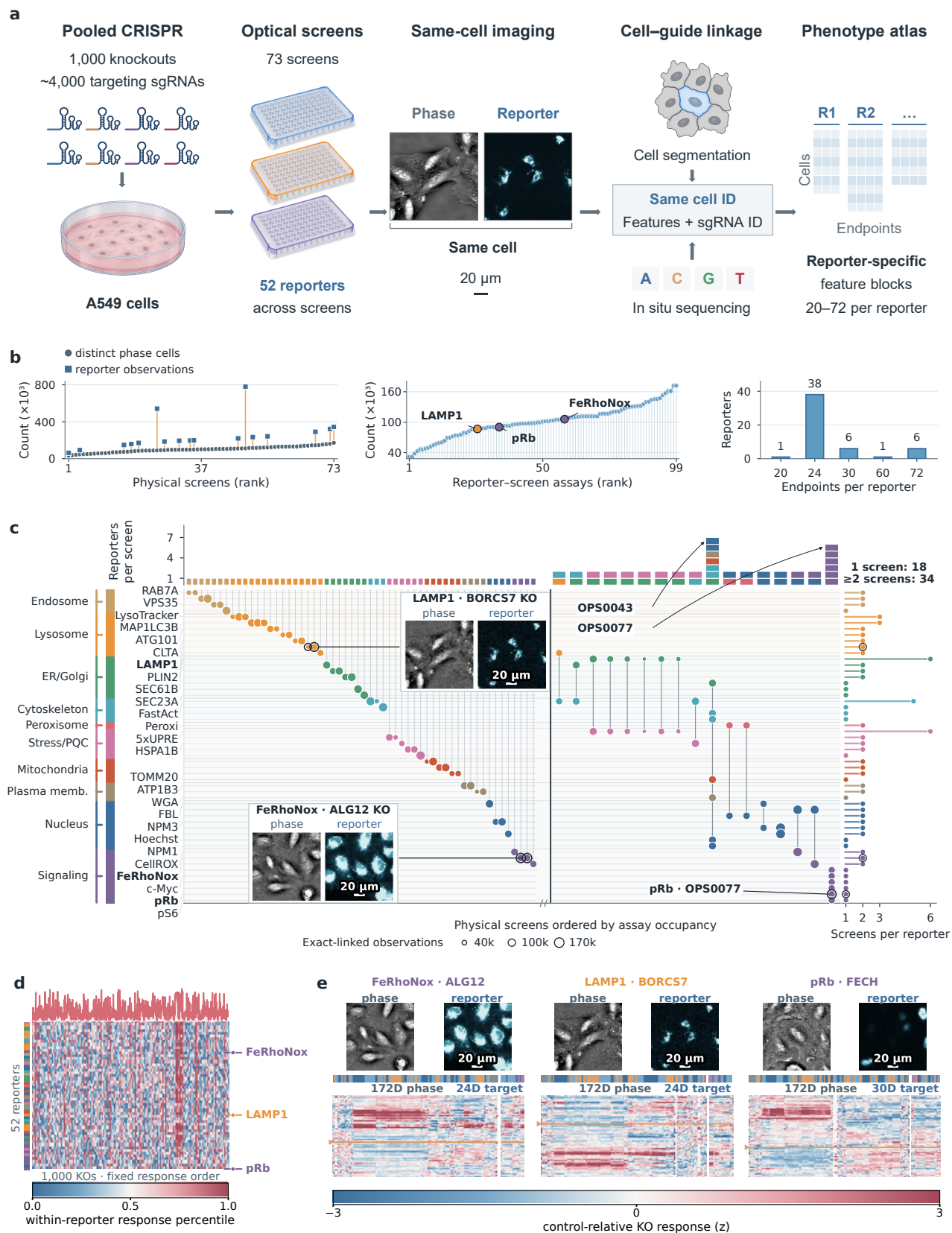
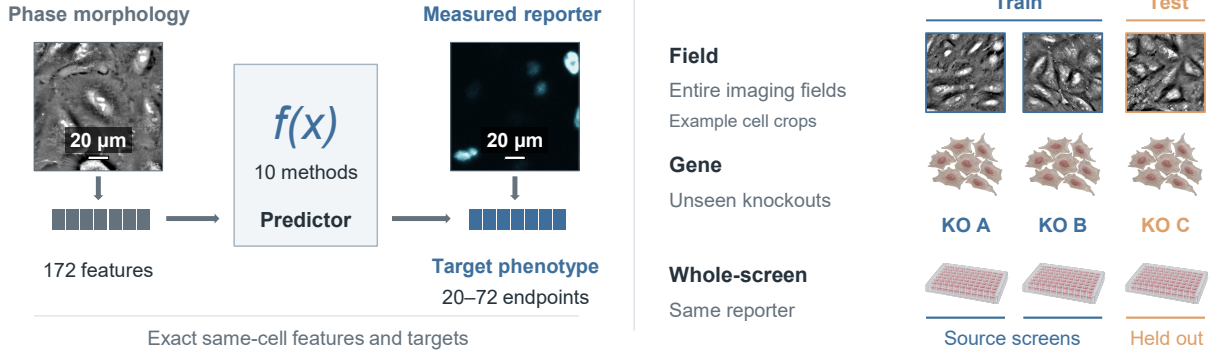
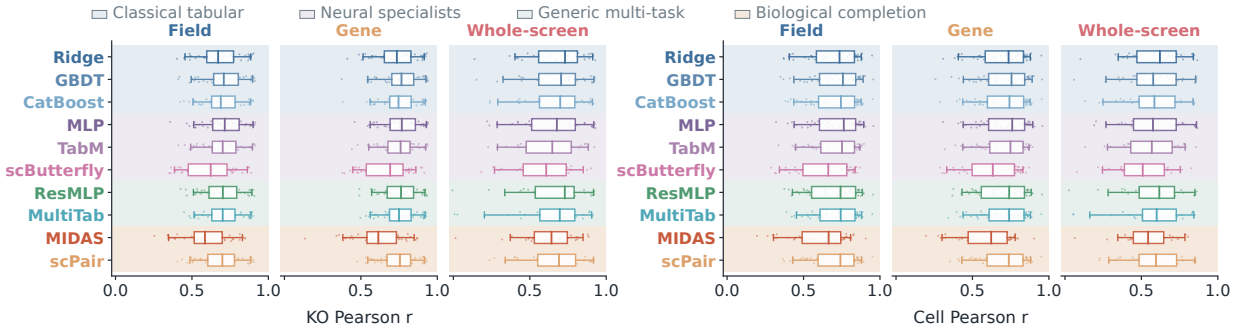


Figure 1: A targeted-measurement atlas with exact same-cell linkage and sparse experimental topology. **a**, Pooled CRISPR imaging and reporter-specific atlas construction. Workflow drawings are schematic; microscopy shows an exact-linked phase–reporter pair. **b**, Left, phase-cell counts and multi-reporter observation totals across 73 screens; middle, coverage of 99 assays; right, reporter frequencies by endpoint-block size. Totals are 7,344,374 distinct phase cells, 9,996,286 phase–reporter observations and 1,604 endpoints. **c**, Complete 52 $\times$ 73 reporter–screen topology with 99 acquired assays; blanks indicate unacquired measurements. Marginal tracks show recurrence, occupancy and observation depth. **d**, Standardized control-relative responses for 1,000 knockouts, converted to within-reporter percentiles and ordered by recoverability for display. **e**, Exact-linked phase/reporter crops and matched knockout-response maps for FeRhoNox, LAMP1 and pRb. Images are outcome-independent examples with reporter-specific contrast, not replicates; scale bars, 20 μ m.

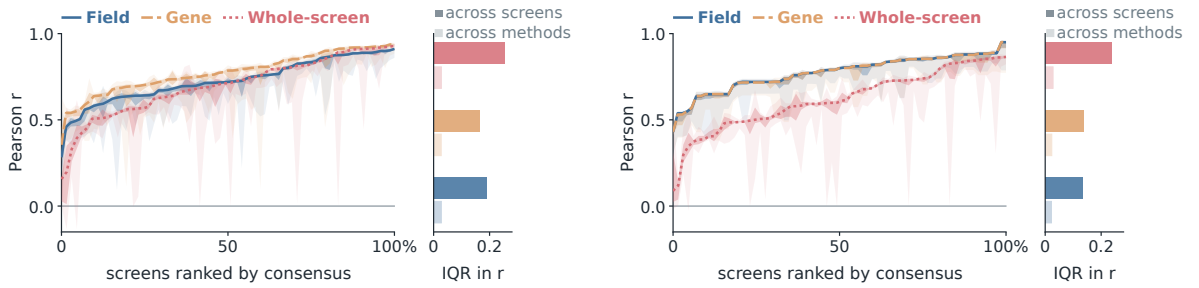
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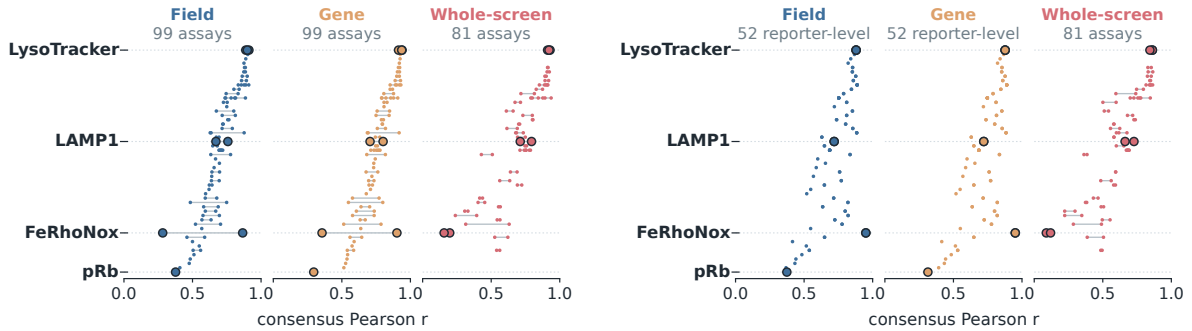


Figure 2: Benchmark performance across generalization, evaluation and aggregation axes. **a**, Same-cell phenotype prediction and schematic field, gene and whole-screen holdouts, with reporter identity retained across screens. Field images are illustrative cell-centred crops, not complete fields or actual partition assignments; scale bars, 20 μ m. **b**, Reporter-level Pearson correlations for ten methods, averaged over five predefined partitions or directed destination-screen evaluations; b–d show knockout responses on the left and cell phenotypes on the right. Points denote reporters; boxes show medians and interquartile ranges, whiskers 5th–95th percentiles and pale bands method families. **c**, Physical screens ranked within each test by the unweighted median across methods, shown as lines with interquartile (dark) and full-range (pale) bands. Adjacent bars compare interquartile variation across screens (solid) and across methods within screens (outlined). **d**, Reporter–screen assays in a shared reporter order; each point denotes an assay and connectors span assays for the same reporter. Field/gene tests contain 52 reporters, 73 screens and 99 assays; whole-screen transfer contains 34 reporters, 66 destination screens and 81 directed assays. Whole-screen cell values are evaluated directly in destination cells. For field/gene cell-level screen and assay views, common per-cell predictions were unavailable for every method; these descriptive panels reweight frozen reporter-level macro-Pearson values by observed screen or assay membership. The 99 assay marks therefore encode 52 distinct reporter-level values; assays for the same reporter have identical cell scores. Reporter-level comparisons use two-sided paired Wilcoxon signed-rank tests against the prespecified MLP, Holm correction across 54 comparisons and an absolute median paired-effect threshold of 0.02. Screen and assay summaries are descriptive.

109 34 repeated reporters, 18 of 66 destination screens and 21 of 81 directed assays (Fig. 3a).

110 Recoverability also followed phenotype content. In the gene test, mean reporter scores ranged from
111 0.877 for endosomal reporters to 0.620 for signalling reporters. Intensity level and intensity extrema or
112 range were the most recoverable endpoint families (means 0.822 and 0.791), whereas spatial displace-
113 ment and sub-object distribution were the least recoverable (0.541 and 0.525; Fig. 3b). Engineered phase
114 morphology was therefore more predictive of global intensity and range than of fine spatial organization
115 in this atlas.

116 We then compared three quantities: how strongly a reporter distinguishes perturbations, how sim-
117 ilar its perturbation structure is to phase, and whether phase predicts its response to an unseen gene
118 (Fig. 3c). Source-study perturbation distinctiveness was modestly associated with held-out-gene recov-
119 erability (Spearman $\rho = 0.344$, 95% bootstrap interval 0.072–0.579; $n = 52$). A single exploratory sub-
120 sampling draw capped acquisition at 300 cells per sgRNA and gave a similar association ($\rho = 0.305$,
121 0.035–0.546), retaining the pattern under that sampling cap. In contrast, phase-reporter overlap was
122 nearly unrelated to recoverability in the directed held-out-gene task ($\rho = 0.090$, -0.174 to 0.345); here
123 overlap denotes undirected similarity between TF-IDF-weighted perturbation-distinctiveness profiles.
124 Thus, a reporter’s perturbation distinctiveness and its similarity to phase provided limited guidance to
125 how well its responses could be predicted for unseen genes.

126 Same-cell morphology carries target information beyond group means and measured covariates

127 Prediction could reflect differences between gene–screen groups even if phase and reporter pheno-
128 types were unrelated among cells within each group. We fixed a prespecified reporter-specific MLP
129 and changed only pairing or phase-feature content across 52 reporters and five held-out-gene partitions
130 (Fig. 4a). Reassigning reporter measurements among cells within training and validation gene–screen
131 groups retained their response distributions while breaking the original same-cell linkage; test cells re-
132 tained their true paired targets. Exact pairing gave median cell- and knockout-level correlations of 0.757
133 and 0.769; within-group target reassignment reduced them to 0.291 and 0.293. Reassignment among
134 cells matched for cell area, density and eccentricity gave 0.531 and 0.502, and size–shape-only prediction
135 0.261 and 0.329; removing size and shape from the 172 features left exact-pair performance essentially
136 unchanged. Mean exact-pair advantages over these three controls were 0.407/0.465, 0.229/0.265 and
137 0.433/0.426 at the cell/knockout levels. Thus, same-cell morphology carried target information beyond
138 group means and the measured covariates.

139 We next predicted variation around each gene–screen mean (Fig. 4b). Within each partition, resid-
140 ual targets used peer-cell condition means excluding the evaluated block. Residual correlations were
141 positive for all 52 reporters (median 0.662), but below exact-cell correlations (0.757). Because the refer-
142 ence means use targeted measurements from other cells in the same held-out condition, this analysis
143 tests conditional within-condition association. Together, the pairing and residual analyses show that
144 morphology contains information about individual-cell targeted states beyond the average response of
145 a perturbation group.

146 Same-cell information did not guarantee quantitative fidelity at the perturbation level. Element-
147 wise median predictions across ten methods showed high rank fidelity and a near-unity magnitude-
148 variance ratio for LysoTracker (magnitude Spearman $\rho = 0.988$; variance ratio 0.930), intermediate fi-
149 delity for ChromaLIVE 488 (0.828; 0.816), and retained rank order but severe variance contraction for
150 pRb (0.708; 0.017; Fig. 4c). The same distinction appeared in strict whole-screen transfer: across the 810
151 method–direction combinations, median knockout-response Pearson was 0.715 and median magnitude-
152 rank Spearman correlation was 0.723, but individual combinations ranged from faithful transfer to
153 marked contraction of the response profile (Fig. 4d). Preserving perturbation order and preserving the
154 spread of response magnitudes are therefore separate capabilities.

155 Target stability, response amplitude, environment and ambiguity mark distinct constraints

156 Low recoverability was associated with several properties of the target and its acquisition. Held-out-
157 gene recoverability correlated with within-screen reliability ($\rho = 0.547$, 95% interval 0.295–0.737; $n = 52$),
158 cross-screen consistency (0.349, 0.014–0.625; $n = 34$) and exact-link coverage (0.482, 0.211–0.686; $n = 52$).
159 Guide-half consistency showed an association of similar direction, but its interval included zero (0.376,

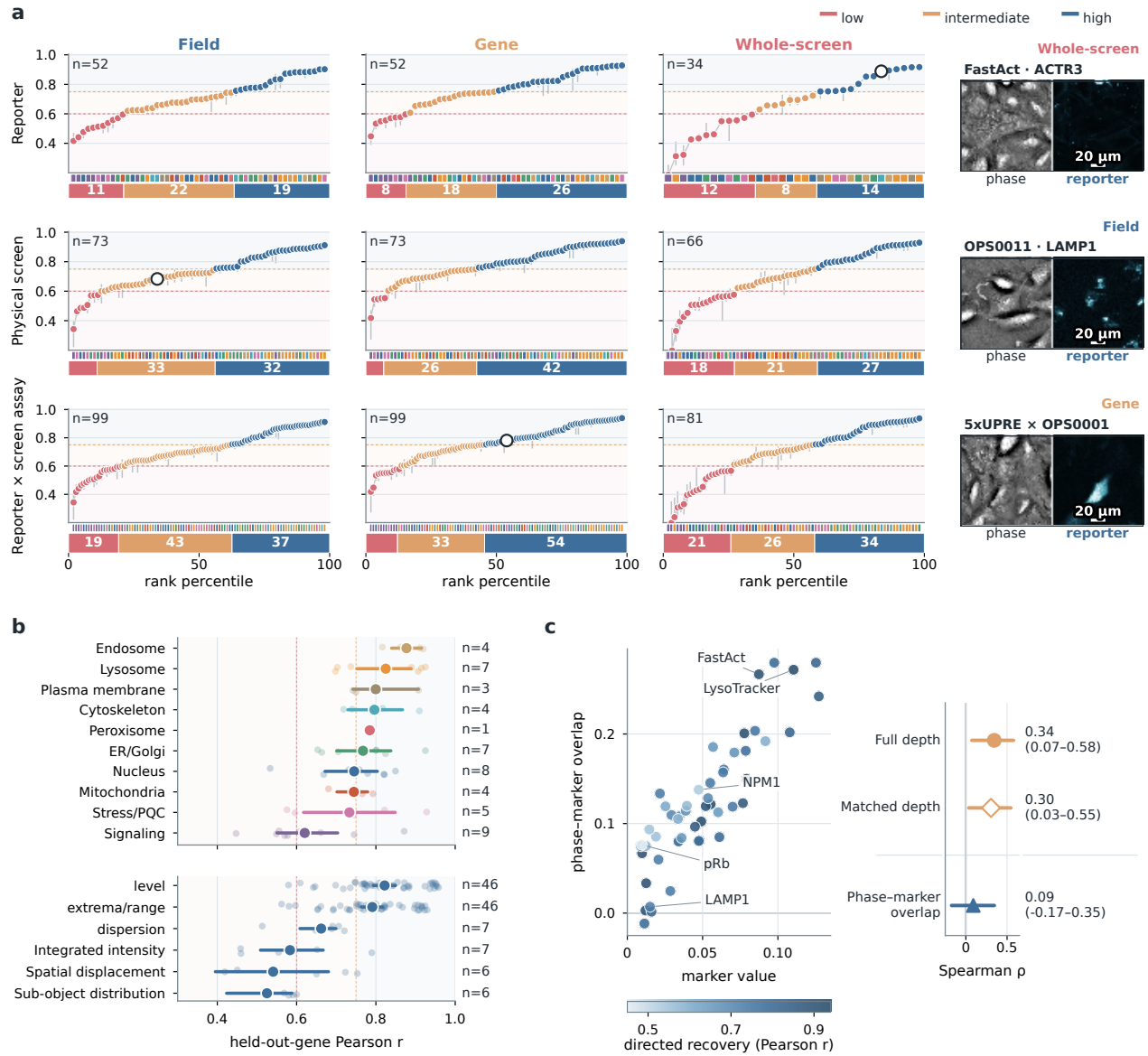


Figure 3: Recoverability is structured across biological and experimental units. **a**, Knockout-response recoverability summarized by the unweighted median across ten methods for reporter, screen and reporter-screen assay units under three tests. Background bands denote low ($r < 0.60$), intermediate ($0.60 \leq r < 0.75$) and high ($r \geq 0.75$) descriptive strata; aligned bars summarize the number of units in each stratum. Sample sizes are 52/73/99 units for field and gene and 34/66/81 for whole-screen transfer. Rings identify three outcome-independent exact-cell callouts shown at right; the images are descriptive and do not enter the statistics. **b**, Held-out-gene recoverability by biological system and endpoint family. Pale points are reporter or reporter-family values; large points are means and lines are reporter-bootstrap 95% intervals from 10,000 resamples. **c**, Reporter value is source-study perturbation-distinctiveness mAP; phase-reporter overlap is Pearson similarity between TF-IDF-weighted perturbation-distinctiveness profiles; matched-budget value uses one source-study subsampling draw with at most 300 cells per sgRNA. Associations with directed held-out-gene recoverability are Spearman correlations across 52 reporters with 3,000 reporter-bootstrap resamples.

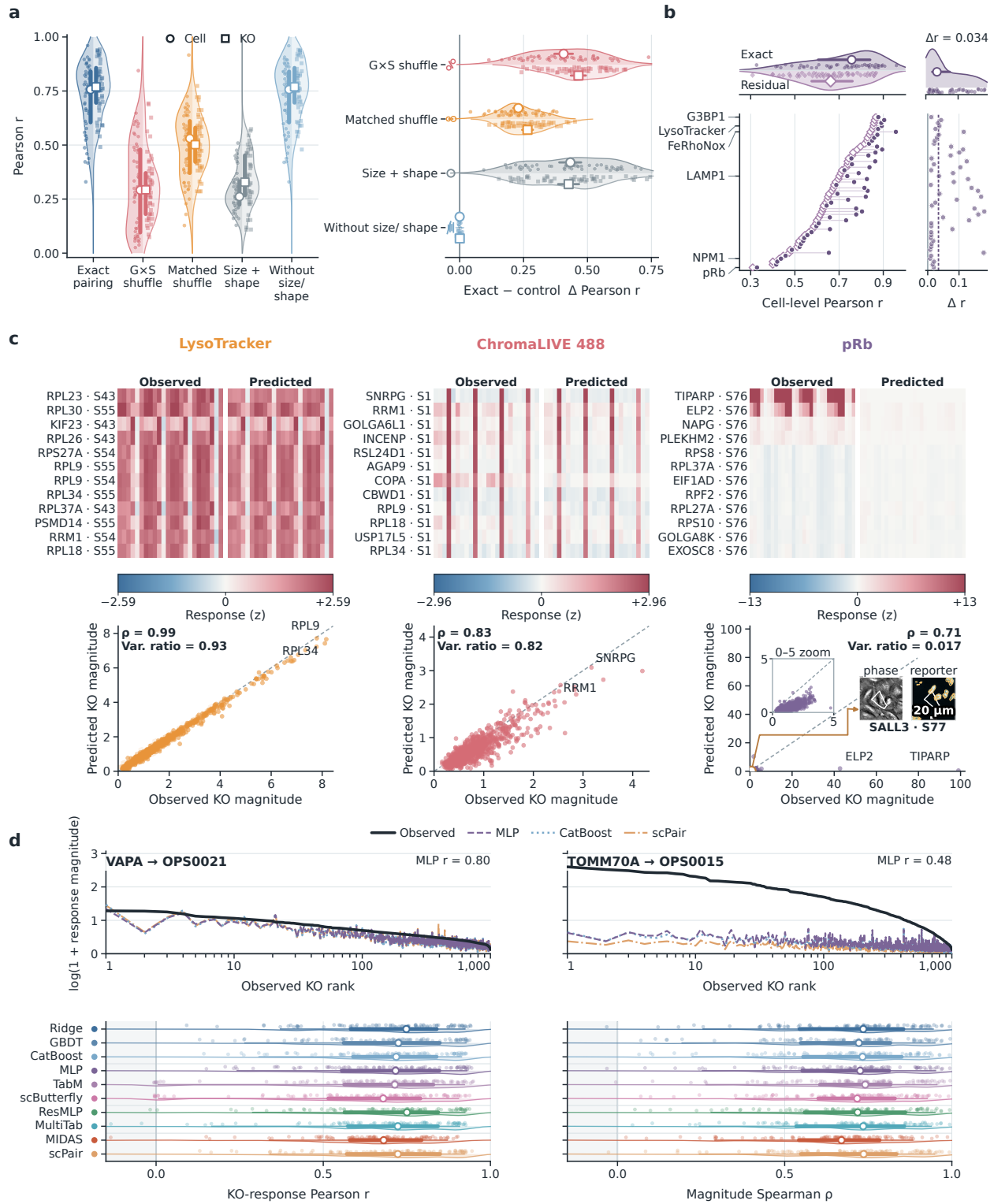


Figure 4: Targeted-state fidelity has separable cell, perturbation and transfer limits. **a**, Reporter-specific MLP recovery under exact pairing and four information interventions ($n = 52$, five held-out-gene partitions); G×S denotes gene-screen shuffling and matched denotes covariate-matched shuffling. Circles/squares denote cell/knockout scores, pale lines pair reporters, and open effect symbols show means with reporter-bootstrap 95% intervals. **b**, Exact-pair prediction versus an MLP fitted to cross-fitted within-gene-screen target residuals ($n = 52$), with ranked differences. Residuals use targeted measurements from peer cells in the held-out condition. **c**, Observed versus median-of-ten-method predictions on a common reporter-wide control reference: 12 strongest held-out gene-screen profiles per heatmap and all 1,000 genes per full-range scatterplot; the pRb zoom shows 0–5 on both axes. The microscopy inset identifies an exact SALL3 cell from OPS0077 (S77); scale bar, 20 μm . **d**, Transfer curves show two prespecified reporter-to-destination directions and three representative predictors. Half-rainclouds retain all 81 directed transfers per method, displaying density, median, interquartile range and 5th–95th percentiles for response reconstruction and magnitude ranking.

160 -0.002 to 0.668 ; $n = 35$; Fig. 5a). In the 35 complete cases, within-screen reliability, guide consistency
161 and log-transformed coverage jointly accounted for 38.7% of the in-sample variation in recoverability
162 ($P = 0.001$ from 3,000 response permutations). The overlapping associations link recoverability to both
163 reproducibility of the measured response and the number of available observations.

164 pRb and NPM1 illustrate why target reproducibility matters for interpreting prediction failure. Me-
165 dian knockout-response agreement across well pairs, guide halves and random cell halves was -0.015 ,
166 -0.020 and 0.158 for pRb, and 0.069 , 0.054 and 0.197 for NPM1 (Fig. 5b). Agreement across wells and
167 guide subsets was lower than agreement across random cell halves. For these targets, the measured per-
168 turbation response itself provided a weakly reproducible reference against which to judge prediction.

169 Contraction of response magnitudes represented a different limitation (Fig. 5c). Across 52 reporters,
170 median magnitude-rank Spearman correlation was 0.825 , predicted-to-observed magnitude-variance ra-
171 tio was 0.779 , top-5% hit recall was 0.790 and within-gene profile Pearson correlation was 0.874 . Eleven
172 reporters retained less than half of the observed variance of knockout-response magnitudes; ten of these
173 eleven nevertheless had magnitude-rank correlations above 0.5 . These reporters preserved much of the
174 perturbation hierarchy while compressing the differences in response magnitude between genes.

175 Variation between screening contexts marked a third constraint. Across 81 directed screen tests,
176 standardized differences in the mean phase and targeted phenotypes of non-targeting controls (NTCs)
177 were associated with loss of transfer performance ($\rho = 0.687$ and 0.751 ; Fig. 5d). After summarizing
178 the directions within each of 34 reporters, the corresponding correlations were 0.765 and 0.837 (both
179 reporter-level permutation $P = 1/3,001$). In two selected FeRhoNox transfer-failure examples, visible
180 changes in the control-cell context accompanied deterioration in prediction. The atlas-wide association
181 links transfer loss to differences in control-cell phenotypes across acquisition contexts.

182 Finally, nearby cells in the engineered morphology space did not always have similar targeted phe-
183 notypes. Within a reporter-screen-gene stratum, the median targeted-state discrepancy of morphology-
184 nearest pairs, relative to eligible reference pairs, was 0.681 and 0.849 for FeRhoNox in two screens and
185 0.889 for pRb (Fig. 5e). A ratio near one means that selecting a morphologically similar neighbour re-
186 moves little of the target-state discrepancy. A raw-thumbnail sensitivity analysis gave ratios near or
187 above one. Residual target variation within narrow morphology neighbourhoods is a plausible source
188 of the averaging and magnitude contraction seen in predictions, although these comparisons do not
189 identify its mechanism.

190 Replacing measured responses reveals which biological analyses are retained

191 To determine which scientific uses survived response replacement, we replaced each reporter's observed
192 responses to held-out perturbations with predictions from the fixed ten-method ensemble and repeated
193 the same downstream analysis (Fig. 6a). The predictors had access to reporter measurements from train-
194 ing perturbations. We assessed recovery of the 50 strongest perturbation responses and overlap between
195 the ten highest-ranked enrichment terms obtained from observed and predicted profiles. The annotation
196 views comprised Gene Ontology biological processes and cellular components^{39,40} and curated protein
197 complexes⁴¹. Median magnitude-rank agreement was 0.825 , top-5% hit recall was 0.790 , and top-ten
198 term overlap was 0.603 , 0.667 and 0.667 for the three annotation views.

199 Hit and ranked-annotation retention were summarized in a cohort-relative utility index, which
200 ranged from 0.084 to 0.964 in the all-reporter display. For predictive evaluation, utility percentiles
201 instead used training-reporter reference distributions (Methods). On this scale, 15 of 52 reporters
202 combined recoverability above the 0.70 criterion with utility below the cohort median, whereas 2
203 showed the converse pattern (Fig. 6b). The median was used only to describe the joint distribution and
204 did not enter the tier rules. Predictive recovery and retention of the selected downstream analyses thus
205 captured different aspects of measurement use.

206 Recoverability, magnitude-rank fidelity, magnitude-variance fidelity and within-screen reliability
207 were associated with retained utility. A Ridge model using these four quantities predicted the utility
208 index in nested leave-one-reporter-out evaluation (Spearman $\rho = 0.786$; $R^2 = 0.587$; Supplementary
209 Figure 4a). For each outer reporter, both the percentile reference distributions and the regularization pa-
210 rameter were determined using the other 51 reporters only. The corresponding univariate associations
211 were 0.690 , 0.791 , 0.635 and 0.659 . Magnitude-rank fidelity had the strongest univariate association,

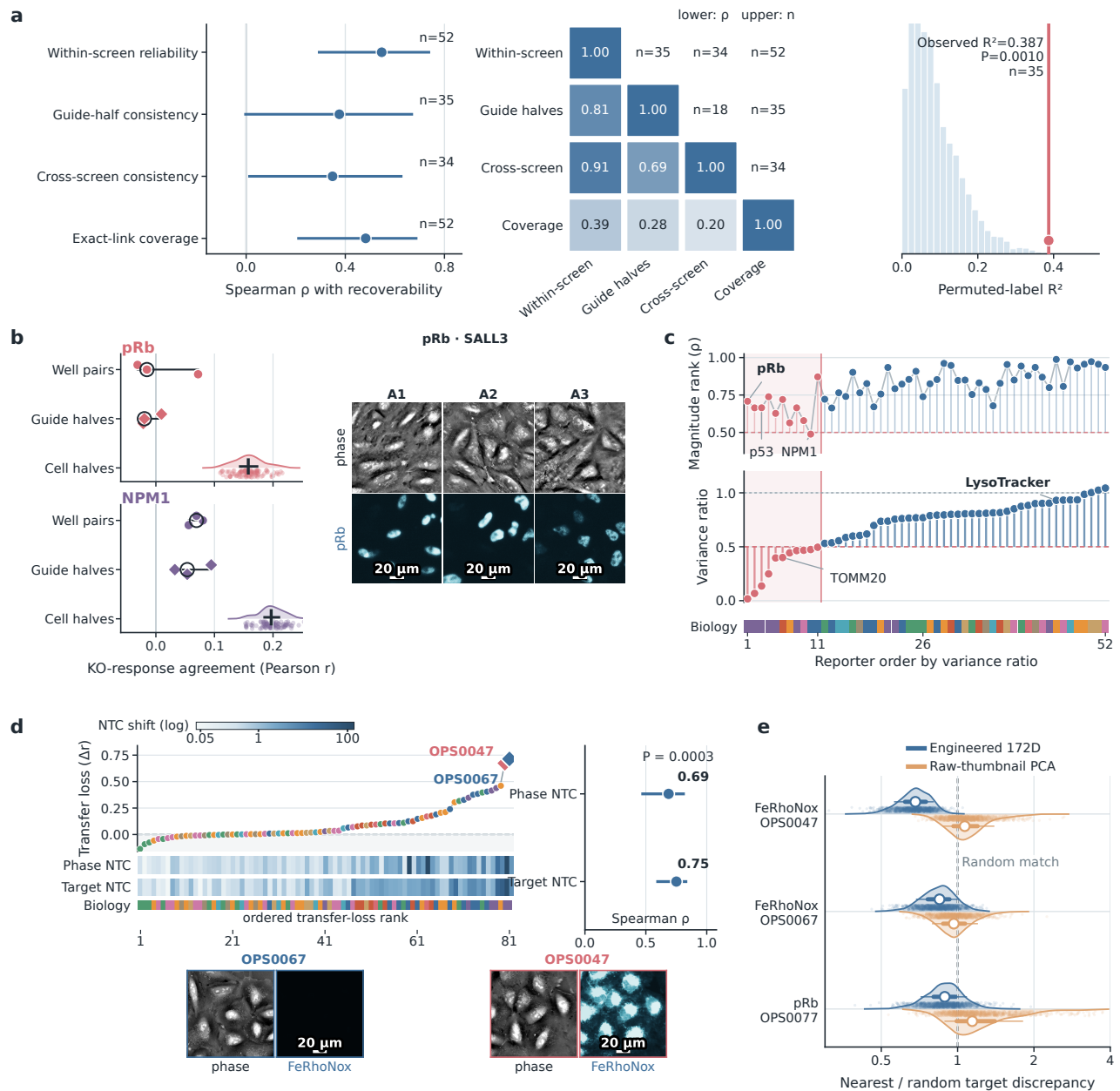


Figure 5: Distinct constraints accompany morphology-based measurement sufficiency. **a**, Recoverability versus within-screen reliability ($n = 52$), guide-half consistency ($n = 35$), cross-screen consistency ($n = 34$) and $\log_{10}$ coverage ($n = 52$): Spearman correlations with 95% intervals from 3,000 reporter-bootstrap resamples; the matrix shows correlations below and sample sizes above the diagonal. The joint model uses 35 complete reporters ($R^2 = 0.387$; $P = 0.001$, 3,000 permutations). **b**, pRb and NPM1 response agreement across three well pairs, three guide-half partitions and 100 random-cell-half draws, with outcome-independent pRb medoids. All values are shown; summaries indicate median and range for well/guide comparisons, and median and interquartile range for cell halves. **c**, Magnitude-rank agreement versus predicted-to-observed magnitude variance for 52 reporters, each containing 1,000 genes and using element-wise median predictions across ten methods. Red highlights 11 reporters retaining less than half the variance; 10 retain rank correlation above 0.5. **d**, Non-targeting-control phase and target shifts versus transfer loss across 81 directions nested in 34 reporters and 66 destination screens. Correlations use directions with reporter-cluster bootstrap 95% intervals; P values use 3,000 reporter-median permutations with plus-one correction, and diamonds identify two predefined FeRhoNox examples. **e**, Target-state discrepancy among morphology-nearest cells relative to eligible pairs, using 172 engineered features or 32-dimensional raw-phase principal components. Engineered-feature comparisons include 992/998 FeRhoNox genes and 1,000 pRb genes, and raw-phase comparisons include 1,000 each; open circles show medians, thick bars interquartile ranges, thin bars 10th-90th percentiles and dashed lines a ratio of one. Biological-system strips accompany **c,d**; scale bars in **b,d**, 20 μm .

consistent with the response-order information shared by perturbation ranking and hit recovery. The multivariable analysis relates fidelity and reliability to downstream retention; it was not a test of incremental benefit over rank fidelity alone.

Atlas-derived rules fixed before final assignment classified 30 reporters as quantitative proxies, 7 as ranking proxies, 3 as requiring direct measurement, 9 as not identifiable because reliability was low or unavailable, and 3 as unresolved (Fig. 6c); domain sensitivity was recorded separately. The quantitative tier identifies reporters meeting the specified criteria for response reconstruction, magnitude ordering, magnitude variance and strong-response recovery. Functional-analysis retention provides a separate assessment. VPS35, for example, met the quantitative response criteria but shared no terms between its observed and predicted top-ten biological-process lists. Thus, response-level agreement can coexist with different leading functional terms for the strongest perturbations. The evidence for each reporter distinguishes these uses rather than assigning a single measure of substitutability. Applications requiring unbiased absolute effect sizes additionally need calibration beyond the magnitude-variance criterion used here. Complete criterion-level evidence for all 52 reporters is shown in Supplementary Figure 1a.

Predictor choice also changed the measurement uses supported for individual targets. Keeping the data partitions, reference reliability and decision thresholds fixed, we recomputed every response criterion for each of the ten predictors. Among the 43 reporters with sufficient reference reliability, 20 supported proxy use under every predictor, 6 did not support it under any predictor and 17 crossed the proxy-use boundary between predictors (Supplementary Data 1). Of the 20 consistently supported targets, 15 changed between quantitative and ranking proxies. For LAMP1, the MLP met the quantitative criteria, MIDAS retained ranking fidelity but compressed response magnitudes, and scButterfly did not meet either proxy tier. Model choice therefore affected the uses supported by a prediction, not only its recovery score.

Using the median score across the ten benchmark methods in each of 81 directed destination-screen evaluations of 34 repeated reporters, recovery remained on the same side of the 0.70 criterion for 30 reporters and magnitude rank for 25; the remaining 4 and 9 reporters, respectively, crossed the criterion between contexts (Supplementary Figure 4b). These changes track the context dependence of two components of the sufficiency assessment.

Exploratory analyses examined how the number of paired reporter measurements and the choice of image representation affected recovery (Supplementary Figure 2). In the 12-reporter label-efficiency analysis, mean recovery improved when the fraction of available paired measurements increased from 0.1% to 1% for all ten methods and further at 20% for nine, across target-only and atlas-assisted training settings. Across six reporters, mean held-out-gene recovery was 0.739 with engineered features, 0.594 with frozen Cytoland and 0.613 after partial fine-tuning. Fine-tuned Cytoland remained below engineered features (paired difference -0.127 , reporter-hierarchical 95% interval -0.188 to -0.066). These results identify the amount of paired training data and the image representation as practical considerations when extending the assessment beyond the full-label, engineered-feature setting. The complete full-label predictor-target decision map and the LAMP1 criterion-level comparison are shown in Supplementary Figure 3.

Independent applications connect prediction fidelity to specific measurement uses

To examine whether the assessment could guide measurement use beyond the atlas, we applied the same response and reliability criteria to two public resources with different biological systems and measurement designs (Fig. 6f). We sought a broad lower-cost input, a targeted readout, paired observations, perturbations, matched controls and repeated target acquisition. Of 59 recorded resources, 42 had enough information for an eligibility assessment and two supported applications using the capabilities they provided (Supplementary Table 1). The survey included harmonized perturbation collections⁴²; incomplete verification and differences in data availability precluded an exhaustive census. OASIS paired brightfield profiles and biochemical measurements at the well level, whereas PERISCOPE used fluorescent predictor channels. For this response-fidelity assessment, numerical thresholds were unchanged and metrics were estimated according to each experimental design (Methods).

The first resource paired brightfield images with two biochemical readouts in primary hepatocytes exposed to 1,085 compounds at eight concentrations⁴³. Using CellProfiler-derived brightfield features,

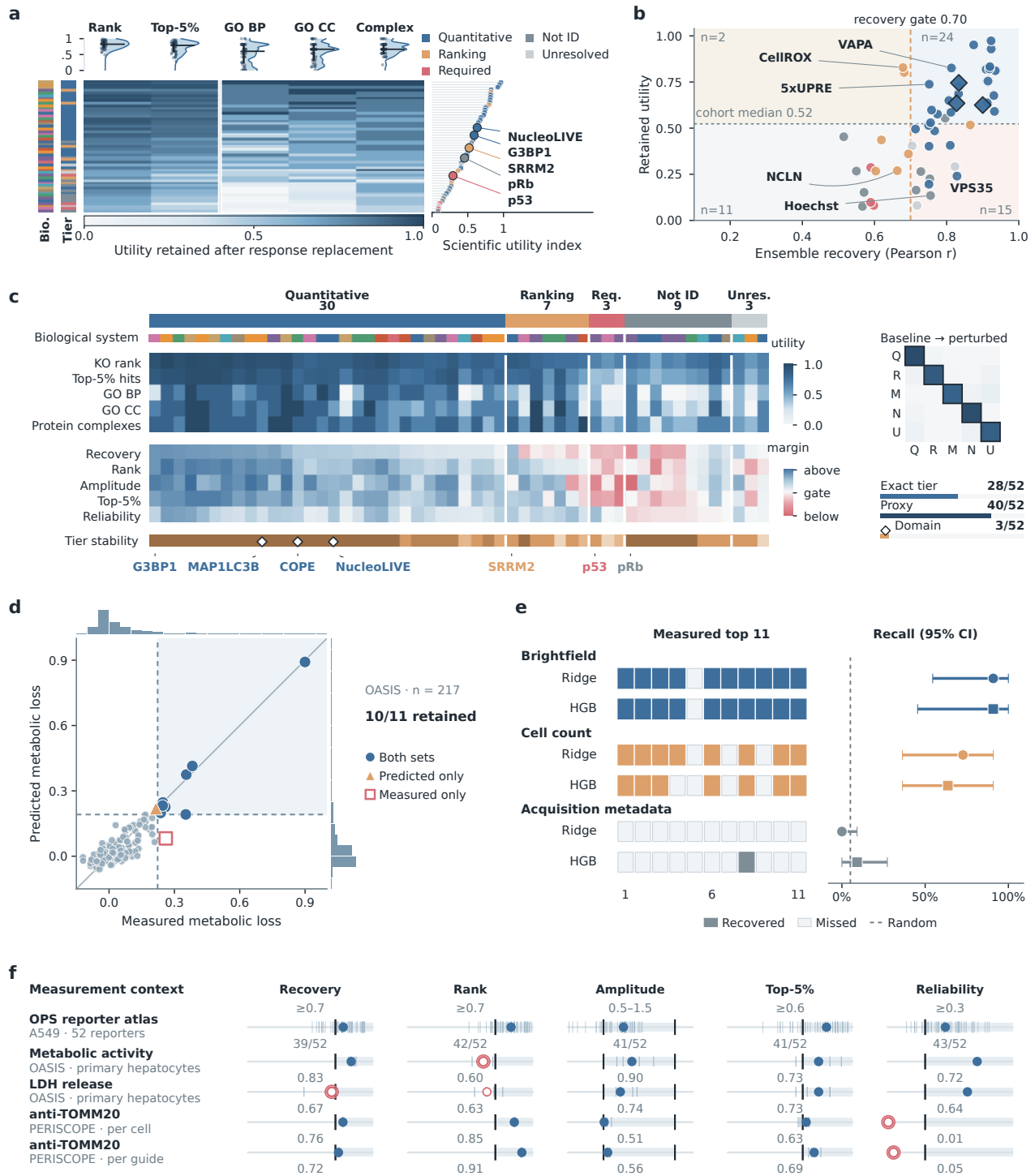


Figure 6: Response fidelity and retained utility support distinct measurement decisions. **a**, Retention after median-of-ten-method response replacement (52 reporters); points, medians and interquartile ranges are shown. Utility combines cohort-percentile hit and functional-term retention. **b**, Ensemble recoverability versus leave-one-reporter-out training-reference utility ($n = 52$); 0.70 is the recovery criterion, whereas median utility is descriptive. Colours indicate tiers and diamonds domain sensitivity. **c**, Retention, five decision margins and tier stability across 36 threshold scenarios; Q/R/M/N/U denote quantitative, ranking, required, not-identifiable and unresolved tiers. **d**, Measured and brightfield-predicted metabolic-activity loss for 217 held-out compounds in primary hepatocytes, with marginal distributions of signed, dose-averaged loss from same-plate DMSO. Dashed boundaries separate top-eleven sets sharing ten compounds; grey points belong to neither set. **e**, Recovery of the same eleven measured hits by Ridge and histogram gradient boosting (HGB) with three input sets: filled cells indicate recovered hits, empty cells missed hits, and columns follow descending measured loss. Adjacent top-5% recalls have 95% compound-bootstrap intervals from 2,000 resamples; cell counts are fluorescence-derived diagnostic inputs and the dashed reference is random selection (11/217). **f**, Response criteria across OPS, OASIS and PERISCOPE: circles show medians or external summaries, ticks individual reporters or replicate evaluations, counts OPS criterion passes and red rings the first failed external criterion. Reliability uses cell halves, dose-averaged well halves and guide halves, respectively; these broad response criteria assess a different use from the signed-loss prioritization in **d**, **e**.

recoverability was 0.83 for metabolic activity and 0.67 for lactate dehydrogenase release. Top-5% hit recall was 0.727 for both readouts, retaining most of the strongest compound responses, but neither met the magnitude-rank criterion for a proxy. Both were unresolved (Supplementary Table 2, Fig. 6f): strongest-hit retention did not establish preservation of the broader response ordering. Dose-averaged split-half reliability estimates were 0.72 and 0.64, above the initial 0.30 criterion. Metabolic activity remained unresolved in all four batches; lactate dehydrogenase release required measurement in batches 25 and 30, where recoverability fell below 0.60 while magnitude-rank fidelity remained below the proxy criterion (Supplementary Table 3). Response fidelity, and hence the supported use of a predictor, therefore varied within the same experimental resource.

We next asked a more specific question: could brightfield predictions prioritize compounds producing the largest decreases in metabolic activity when only a small subset could receive follow-up measurement? Using the resource's released brightfield-channel DINO representations, we fitted a fixed Ridge predictor on 868 development compounds and evaluated 217 held-out compounds, with all measurements of the test compounds excluded from fitting across doses and production sources. The ranking score was the signed mean loss of RealTime-Glo metabolic-activity signal relative to same-plate DMSO controls, averaged over each compound's eight assayed concentrations. Selecting the predicted top 5% recovered 10 of the 11 compounds with the largest measured losses (90.9%; compound-bootstrap 95% interval, 54.5–100%; Fig. 6d,e). The selected compounds had a mean measured loss of 0.334 normalized signal units, compared with 0.338 for the measured top eleven, indicating retention of substantial decreases rather than reordering of near-zero responses.

To evaluate this use against repeated measurement, we identified 94 held-out compounds with all eight concentrations matched across two production sources. Predictions from one source recovered four of the five strongest measured losses in the other source, in both directions (Supplementary Figure 4c). Measured-source-to-measured-source prioritization also recovered four of five. The source roles comprised source 26 paired with sources 27 or 30; they were not new sources held out from training. The paired recall-difference intervals were wide, so the matching point estimates establish a useful repeat-measurement reference rather than equivalence. A fixed gradient-boosting predictor also recovered 10 of 11 in the full held-out set; cell-count and acquisition-metadata comparators provide diagnostic references for the same measured-hit set (Fig. 6e). Within-plate disruption of prediction-compound correspondence reduced the Ridge mean recall to 5.38%. Larger selection budgets and source-pair-specific results are retained in Supplementary Figure 4c,d and Supplementary Data 3.

This prioritization analysis evaluates signed metabolic-activity loss at a fixed selection budget, using brightfield DINO features and dose-matched aggregation. It complements the broader CellProfiler-based response assessment by testing the selection of compounds for follow-up directly.

The second resource used four Cell Painting dyes to predict an anti-TOMM20 channel in A549 cells and was evaluated both per cell and per guide profile¹¹. Fold-median prediction metrics met every response-fidelity threshold at both levels; at the guide-profile level, all five held-out-gene folds individually met these thresholds. Correlations between disjoint sgRNA partitions were 0.005 and 0.047, below the reliability criterion of 0.30. This reference measures consistency between guide-defined perturbation responses, combining assay variation, guide efficacy and biological heterogeneity. Restricting the guide-level analysis to core essential or mitochondrial genes gave reliability values of 0.069 and 0.051, respectively, leaving both below 0.30 (Supplementary Table 4). The classification remained not identifiable: the available profiles supported recovery of the measured target responses but did not establish their reproducibility across independent guide sets.

A within-plate input-permutation control reduced both CellProfiler-based OASIS predictions, for metabolic activity and LDH release, to the measurement-required category. In PERISCOPE, input permutation reduced prediction fidelity while leaving the not-identifiable classification unchanged because target reliability was assessed first (Supplementary Figure 1 and Supplementary Note 1). The two controls test different parts of the assessment: prediction fidelity determines a decision only when the reference response is sufficiently reproducible. Predicting each Cell Painting dye from the other three also yielded not-identifiable assignments for all four dyes (Supplementary Figure 1 and Supplementary Note 2). Reliability was below 0.30 across dyes despite differing prediction fidelity, identifying a shared limitation of the available perturbation-response evidence.

317 Together, the external applications extend measurement-sufficiency assessment to well-paired bio-
318 chemical readouts and fluorescent perturbation profiles. They distinguish retained strongest-hit in-
319 formation from broader response ordering in OASIS, and cross-channel response fidelity from guide-
320 partition reproducibility in PERISCOPE, allowing the measurement decision to follow the evidence
321 available for each biological task.

322 Discussion

323 Morphology-based prediction preserved different properties of targeted cellular measurements to differ-
324 ent degrees. Exact same-cell controls identified target-specific information beyond perturbation-group
325 means and measured covariates, yet perturbation ranking could remain accurate when the spread of
326 response magnitudes contracted. Response-level agreement also differed from retention of functional
327 analyses. These findings connect the promise of virtual-cell models to a concrete measurement question:
328 which aspects of a targeted readout are recoverable from an observed cellular state, and which scientific
329 uses still depend on direct measurement?

330 Method choice is consequential within this framework. Paired reporter comparisons supported gains
331 for the MLP over several comparators, while the apparent ordering also depended on generalization
332 test and reporting unit. Recomputing the measurement criteria for each predictor linked these perfor-
333 mance differences to supported uses: some targets remained proxies across all ten methods, whereas
334 others changed their quantitative, ranking or proxy-use status. Model development can therefore ex-
335 pand measurement capability, but its success must be judged for the target and intended use rather than
336 by correlation gains alone. The fixed ensemble provides one assessed prediction rule, not an intrinsic
337 classification of the reporter.

338 The distinction between response fidelity and scientific use is central to measurement sufficiency. The
339 operational criteria identified 30 quantitative and 7 ranking proxies, while the VPS35 example showed
340 that a quantitative response-level assignment can coexist with loss of the leading biological-process
341 terms. Reporting hit recovery and functional-term retention alongside the response metrics makes this
342 difference explicit. The cohort-relative utility index summarizes those selected analyses; its interpreta-
343 tion remains tied to the tasks included and the reporter population used as a reference. This extends
344 application-appropriate validation from image-analysis accuracy to the biological analyses performed
345 after a measurement is replaced²⁷.

346 The observed limitations point to different ways to improve prediction and measurement design.
347 Poor agreement between well or guide partitions makes reference-response reproducibility a priority.
348 Screen-dependent performance makes validation in the intended acquisition context important. Differ-
349 ences in reporter phenotypes between morphologically similar cells motivate richer image representa-
350 tions or additional measurements, because ambiguity in the present feature space need not imply ab-
351 sence of the information from the image. These associations guide the choice of follow-up experiments
352 without resolving the underlying causes.

353 In practice, the intended biological decision determines which evidence is needed. A paired pilot
354 can estimate target reliability and prediction fidelity under a holdout that matches the proposed use,
355 followed by evaluation of the actual downstream analysis. The present response criteria provide start-
356 ing points for perturbation prioritization and response comparison; tolerances should reflect the conse-
357 quences of error in the intended assay. Targeted training measurements remain necessary in this study,
358 so the proxy assignments support prediction for new perturbations after paired data have been acquired.
359 Extending these uses to a new cell system or acquisition context requires new validation.

360 The external applications extend this use-specific interpretation to different assay designs. Bright-
361 field predictions in primary hepatocytes supported prioritization of the largest metabolic-activity losses
362 on held-out compounds, with repeated measurements providing a reference for allocating follow-up
363 measurements. Cross-channel prediction of anti-TOMM20 was instead constrained by reproducibility
364 across guide partitions. Because cell halves, replicate wells and guide partitions capture different sources
365 of variation, assessment in a new system requires a reliability estimate and acceptance criteria matched
366 to its experimental design and intended use.

367 The primary evidence comes from A549 cells in one atlas, with strict screen-transfer tests available
368 for 34 reporters. The engineered 172-dimensional representation may omit spatial information, and

the six-reporter image comparison covers only a small subset of possible alternatives^{44–46}. Thresholds were selected on this atlas and require calibration for new intended uses. Across 36 local variations of these thresholds, 40 of 52 reporters remained on the same side of the proxy boundary and 28 retained the same tier, with most movement between quantitative and ranking proxies (Supplementary Table 5). Measurement sufficiency is consequently a property of a target–predictor–context combination assessed for a specified scientific use. Preserved perturbation ranking, faithful response magnitudes and retained functional interpretation are distinct outcomes, each of which can support a different measurement decision.

Methods

Study design, data resources and phenotype representations

We analysed the public optical perturbation atlas of A549 cells³⁰; no new biological samples were generated. The source study combines pooled CRISPR knockouts, phase and targeted-fluorescence imaging, perturbation identities obtained by in situ sequencing and engineered single-cell phenotypes. The analysed data comprised 1,000 knockout genes, 52 reporters, 73 physical screens and 99 observed reporter–screen assays spanning ten biological systems. Quality-controlled linkage yielded 7,344,374 distinct phase cells and 9,996,286 exact-linked phase–reporter observations because one phase cell could contribute more than one reporter observation. Reporter–screen combinations that were not acquired were treated as missing assays, never as negative measurements.

Reporter was the primary inferential unit for atlas-wide claims. Physical screen and reporter–screen assay were additional reporting units; genes, endpoints, cells and data partitions were nested observations. Dependence among directed whole-screen tests was handled using reporter-level summaries and reporter-preserving resampling, as specified below. Inclusion required a valid source linkage, an observed reporter endpoint block and assignment to a prespecified partition. Exclusions and unresolved mappings are reported in the deposited panel-level source tables.

Phenotype representations. The primary input was a 172-dimensional engineered morphology vector derived from each segmented phase image. The features described image intensity, texture, object geometry and spatial organization. The target was a reporter-specific engineered phenotype derived from the fluorescence image of the same cell. The target schema comprised one 20-dimensional block, 38 24-dimensional blocks, six 30-dimensional blocks, one 60-dimensional block and six 72-dimensional blocks, for 1,604 endpoints in union. Fitting and evaluation used only the endpoints observed for a reporter; heterogeneous blocks were not padded with synthetic targets.

Model preprocessing was fitted within the training portion of each partition. Phase variables finite in fewer than 80% of training cells were removed; remaining non-finite values were replaced by the training median, variables with training standard deviation no greater than 10^{-8} were removed, and retained variables were centred and scaled by their training means and standard deviations. A cell was excluded for a reporter if any endpoint in that reporter’s observed target block was non-finite. Target endpoints were standardized by reporter using training means and standard deviations, with invariant endpoints removed. The fitted feature filters, imputation values and scaling parameters were applied unchanged to validation and test cells. Size and shape features were identified from the feature schema before the same-cell controls were evaluated. Cell, gene, guide, field, screen and reporter identifiers were retained in every prediction table to reconstruct the evaluation units independently of phenotype values.

Prediction models and generalization framework

Three prespecified evaluation regimes represented distinct deployment questions. For held-out fields, the physical screen–well–tile identifier was assigned once in the global phase table and hashed into five folds. The same physical field, and any exact-linked phase row reused by different reporter heads, therefore had one fold assignment throughout specialist and shared-model training; knockout identities and screen context remained represented in training. For held-out genes, knockout genes were partitioned into five folds and every cell carrying a test gene was excluded from model fitting and model selection.

This test evaluates response reconstruction for an unseen perturbation within the observed reporter-screen structure. In both fivefold analyses, fold k was the outer test set, fold $(k + 1) \bmod 5$ was the validation set and the other three folds were used for training.

For strict whole-screen transfer, all observations from one physical destination screen were removed from training. A valid direction required the same reporter in at least one source screen and in the held-out destination. This produced 81 directed evaluations across 34 repeated reporters and 66 destination screens. Held-out-field and held-out-gene summaries included 52 reporters, 73 physical screens and 99 reporter-screen assays; whole-screen summaries included 34 reporters, 66 destination screens and 81 directed assays. Reporter scores were arithmetic means over the five partitions for field and gene tests and over eligible destination-screen directions for whole-screen transfer.

Prediction methods. The ten methods were selected before the final comparative analysis: Ridge regression, gradient-boosted decision trees, CatBoost³², a reporter-specific MLP, TabM³³, scButterfly³⁴, a shared residual MLP, MultiTab, MIDAS³⁵ and scPair³⁶. The set covers conventional target-specific tabular prediction, neural tabular models, shared multitask prediction and single-cell multimodal completion. Every method received the same 172-dimensional input, observed reporter-specific endpoints and train, validation and test partitions; training scope, loss functions and model selection are summarized in Supplementary Table 6.

Target-specific methods were fitted separately for each reporter. Shared methods used a common representation with reporter-aware outputs or endpoint masks, and their loss was averaged over observed endpoints only. Hyperparameters and stopping rules were selected on development data and fixed before final evaluation. All downstream analyses used out-of-fold predictions. The benchmark used the validation-selected single-model states; its recoverability landscapes were summarized by the unweighted median of ten method-level scores. The response-replacement ensemble instead combined predictions from the archived parameter-averaged checkpoints of TabM, scButterfly, ResMLP, MultiTab, MIDAS and scPair, with the same fitted Ridge, GBDT, CatBoost and MLP predictors as in the benchmark. Each averaged checkpoint supplied one model prediction; the method-specific states and input files are identified in Supplementary Data 1. For each reporter, fold and gene-screen row, we took the element-wise unweighted median of the ten aligned response vectors before averaging across screens. Held-out-gene fidelity, measurement-sufficiency classification and utility-prediction analyses evaluated this fixed ensemble; their recoverability was calculated directly from its predictions using Eq. 2, followed by the arithmetic mean over five folds. No method received performance-dependent weighting, and methods were not treated as biological replicates.

Published comparator implementations were adapted to accept the common phase input, heterogeneous reporter blocks and predefined partitions. Unobserved reporter endpoints were masked from both the loss numerator and denominator rather than encoded as zeros. Model selection used validation outcomes only. Supplementary Table 6 specifies the architecture, objective, training scope and selection procedure used for each implementation.

Recoverability metrics and aggregation units

Let I_{rsg} denote exact-linked cells for reporter r , screen s and knockout gene g , and let I_{rs0} denote non-targeting controls from the same reporter and screen. For endpoint e , the control-relative response was

$$\Delta y_{rsge} = \frac{1}{|I_{rsg}|} \sum_{i \in I_{rsg}} y_{ire} - \frac{1}{|I_{rs0}|} \sum_{i \in I_{rs0}} y_{ire}, \quad (1)$$

with $\Delta \hat{y}_{rsge}$ defined analogously from out-of-fold predictions. Cell-level recoverability was calculated separately for every observed endpoint across exact-linked test cells and then averaged equally across endpoints. Knockout-level recoverability was the corresponding macro-endpoint statistic,

$$R_r^{\text{KO}} = \frac{1}{|E_r|} \sum_{e \in E_r} \text{corr}_p(\Delta y_{r..e}, \Delta \hat{y}_{r..e}), \quad (2)$$

where E_r is the observed endpoint set for reporter r and the correlation runs over eligible screen-gene response rows in the relevant test partition. A Pearson value required at least two paired observations

and non-zero observed and predicted variance; otherwise that endpoint was undefined and excluded from the equal-endpoint macro-average. A screen-gene response could contain one or more targeting cells but required matched test controls from the same screen. Non-targeting controls define Eq. 1 but are excluded from the cell-level correlation. Scores were then summarized by reporter, physical screen or reporter-screen assay, corresponding respectively to target recoverability, environmental recoverability and recoverability of a particular target in a particular environment. For held-out-field and held-out-gene cell-level screen and assay views, common per-cell predictions were unavailable for every shared method; these views therefore reweighted reporter-level cell macro-Pearson values by observed screen or assay membership. Strict whole-screen cell scores were evaluated directly in destination cells.

Reporter-level method comparisons used a prespecified reporter-specific MLP reference and two-sided paired Wilcoxon signed-rank tests over reporters. A difference was called statistically and practically distinct only when the Holm-adjusted $P < 0.05$ across 54 comparisons and the absolute median paired effect was at least 0.02. Tests paired 52 reporters in the field and gene views and 34 repeated reporters in whole-screen transfer. Screen and assay summaries were descriptive because their units share reporters, genes and physical acquisition structure. Descriptive low-, intermediate- and high-recoverability strata were defined as Pearson $r < 0.60$, $0.60 \leq r < 0.75$ and $r \geq 0.75$, respectively. These strata summarize the recoverability distribution and were not used as measurement-sufficiency tiers.

Biological and phenotype summaries. Reporters were assigned to ten biological systems using source metadata. Endpoints were assigned to six families: intensity level, intensity extrema or range, intensity dispersion, integrated intensity, sub-object distribution and spatial displacement. Biological-system means used reporter as the unit. Endpoint-family means first averaged endpoints within reporter and family; reporter-family values were then summarized. Intervals were estimated by reporter-level bootstrap resampling with fixed random seeds.

Reporter value was defined from the source study's perturbation-distinctiveness analysis. For each mapped targeted marker, we averaged copairs mean average precision (mAP) across 1,001 perturbation labels (1,000 gene knockouts and NTC). Guide-level profiles generated by different sgRNAs for the same perturbation were positives and profiles from other perturbations were negatives, using cosine distance. To quantify undirected phase-marker overlap, we formed a $53 \times 1,001$ matrix containing phase and the 52 mapped marker distinctiveness profiles. For profile r and perturbation g , term frequency was $TF_{rg} = M_{rg} / \sum_h M_{rh}$; document frequency was the number of profiles for which M_{rg} exceeded the across-profile mean for perturbation g ; and $IDF_g = \log\{(53 + 1) / (df_g + 1)\}$. Overlap was the Pearson correlation across perturbations between the TF-IDF-weighted phase and marker profiles. Directed recoverability was the unweighted median across ten methods of held-out-gene knockout-response Pearson. As a sampling-depth sensitivity analysis, we used one source-study subsampling draw that retained at most 300 cells per sgRNA without replacement, re-aggregated guide-level profiles and recomputed perturbation-distinctiveness mAP. All 52 reporters mapped one-to-one to both source tables. Associations were Spearman correlations across reporters; 95% intervals used 3,000 reporter-level bootstrap resamples. Reporter value and cross-modal overlap were kept separate from recoverability because neither is a directed prediction metric.

Same-cell information and falsification controls

The primary same-cell analysis fixed the reporter-specific MLP and changed only the information available during fitting. The design covered 52 reporters, six intervention arms and five held-out-gene partitions, yielding 1,560 reporter-arm-partition results. Reporter was the inferential unit.

Exact pairing used the registered phase vector and targeted phenotype from the same cell. For gene-screen derangement, target vectors in the training and validation sets were reassigned without fixed points among cells from the same gene and screen; the held-out test set retained its true exact pairing. The model was therefore evaluated on the same exact-linked test outcome after its access to same-cell correspondence during fitting had been removed. For covariate-matched derangement, cells were matched without replacement within each gene-screen group by greedy nearest-neighbour pairing in standardized log cell area, log field-tile cell density and eccentricity. Density was the number of exact-linked cells in the same screen, well and imaging tile. These three outcome-independent phase covariates were

514 scaled over the reporter cohort solely to construct the matching control; this transductive distance nor-
 515 malization used no targeted phenotype. Candidate pairs were ordered deterministically by distance and
 516 exchanged their target vectors. When an odd cell remained, it and the nearest existing pair were reas-
 517 signed in a randomly oriented three-cycle; singleton groups were left unchanged. Training and valida-
 518 tion partitions were deranged independently, whereas test cells retained exact pairing. Size-shape-only
 519 prediction restricted input to gross geometry features, whereas phase-minus-size-shape removed those
 520 features from the complete 172-dimensional representation.

521 The residual analysis addressed a different estimand. Within the training, validation and test par-
 522 titions separately, a fixed-seed random permutation divided each gene-screen stratum into $\min(5, n)$
 523 near-equal blocks. For cell i in block b , the target residual was

$$\tilde{y}_{ire} = y_{ire} - \frac{1}{|I_{r,g(i),s(i)}^{(-b)}|} \sum_{j \in I_{r,g(i),s(i)}^{(-b)}} y_{jre}, \quad (3)$$

524 where the reference mean used cells in the same stratum but outside block b ; singleton strata were as-
 525 signed zero residual. The MLP was fitted directly to these residual targets. Because defining a test
 526 residual uses target measurements from peer cells in the held-out test stratum, this arm is a conditional
 527 association analysis rather than a deployable out-of-distribution prediction task; knockout-level recov-
 528 ery is consequently undefined.

529 Each reporter-arm score was averaged over the five held-out-gene partitions. Paired differences were
 530 calculated within reporter, and reporter-resampled intervals were used for mean exact-pair advantages.
 531 The residual arm was evaluated only at the cell level.

532 Fidelity limits and sources of recoverability failure

533 For endpoint-wise recoverability, fold-specific training-set standardization does not affect Pearson cor-
 534 relation. Quantities that combine endpoint coordinates across held-out folds require a single reference
 535 scale. For reporter endpoint e , we therefore defined a common control scale from all unique non-
 536 targeting-control cells after removing the physical-screen-specific control mean,

$$c_{re} = \left\{ \frac{1}{N_{r0}} \sum_s \sum_{i \in I_{rs0}} (y_{ire}^{\text{raw}} - \bar{y}_{rs0e}^{\text{raw}})^2 \right\}^{1/2}. \quad (4)$$

537 Here $N_{r0} = \sum_s |I_{rs0}|$. The common reference used control measurements only and was applied for eval-
 538 uation, independently of model fitting and knockout outcomes. All 1,604 endpoint scales were finite
 539 and exceeded 10^{-8} . If endpoint e had been standardized by training-fold standard deviation $\sigma_{re}^{(f)}$, its
 540 control-relative response was converted as $\Delta y^{\text{common}} = \Delta y^{(f)} \sigma_{re}^{(f)} / c_{re}$, and predictions were converted
 541 identically. All ten methods shared the same fold-by-reporter target transforms. The original fold-
 542 standardized calculation was retained as a scale sensitivity analysis (Supplementary Methods).

543 For each reporter and held-out gene, common-scale control-relative response vectors from different
 544 screens were then averaged with equal screen weight. Observed and predicted response magnitudes
 545 were

$$M_{rg} = \|\Delta \bar{y}_{rg}\|_2, \quad \hat{M}_{rg} = \|\Delta \hat{y}_{rg}\|_2. \quad (5)$$

546 Magnitude-rank fidelity was $\text{corr}_S(M_{rg}, \hat{M}_{rg})$, and the amplitude variance ratio was

$$Q_r^{\text{var}} = \frac{\text{Var}_g(\hat{M}_{rg})}{\text{Var}_g(M_{rg})}. \quad (6)$$

547 Strong-hit recall was the fraction of the 50 genes with largest observed magnitude recovered among the
 548 50 genes with largest predicted magnitude. Profile fidelity was Pearson correlation between observed
 549 and predicted endpoint vectors within gene; it required at least three finite endpoint coordinates with
 550 non-zero variance and was otherwise undefined.

551 Reporter-level metrics were calculated from the element-wise unweighted median of ten aligned
 552 out-of-fold prediction profiles. The LysoTracker, ChromaLIVE 488 and pRb examples were selected

553 to represent quantitative recovery, partial-pattern recovery and rank-preserving amplitude contraction,
 554 respectively, before final visualization.

555 **Target stability and coverage.** Within-screen reliability used a deterministic hash of the phase-row
 556 identifier to split exact-linked cells into disjoint halves. Within each screen and half, reporter endpoints
 557 were robustly standardized to the non-targeting controls using the median and median absolute deviation.
 558 Endpoint responses were averaged by knockout and reduced to a root-mean-square response
 559 magnitude; Pearson correlation between halves was calculated over at least 20 common knockouts. Re-
 560 porter reliability was the median correlation across its screens, and intervals used 1,000 gene-bootstrap
 561 resamples. Guide-half consistency used three predetermined partitions for knockouts represented by
 562 at least four guides. Endpoints were robustly standardized within well, aggregated with equal weight
 563 first across each guide subset and then across wells, and compared as knockout response magnitudes; a
 564 matched null used 100 random cell-half partitions within gene and well. Cross-screen consistency corre-
 565 lated complete-screen knockout-magnitude vectors for every pair of screens sharing a reporter and took
 566 the median pairwise correlation; it was estimable for 34 repeated reporters. Exact-link coverage was the
 567 number of linked reporter observations and was $\log_{10}$ transformed for modelling.

568 Bivariate associations used every reporter with the relevant measurement. A prespecified complete-
 569 case linear model included within-screen reliability, guide-half consistency and $\log_{10}$ exact-link coverage
 570 ($n = 35$). Cross-screen consistency was reported separately because it was available only for repeated
 571 reporters. The in-sample R^2 was evaluated against 3,000 permutations of recoverability values across re-
 572 porters while the three predictors and complete-case set were held fixed. Because the predictors overlap
 573 biologically and statistically, they were interpreted jointly as stability constraints rather than indepen-
 574 dent causal effects.

575 For pRb and NPM1, response agreement was computed for three well pairs, three deterministic
 576 guide-half partitions and 100 random-cell-half partitions. Random-cell splits assess the sampling stabil-
 577 ity of aggregated responses within the observed population. Well and guide comparisons assess agree-
 578 ment across wells and perturbation reagents, respectively, combining sampling, biological and technical
 579 variation. The primary repeat comparison should match the intended use; alternative splits provide
 580 complementary evidence rather than isolate individual noise sources. Registered pRb microscopy used
 581 outcome-agnostic medoid cells from the same guide across three wells and served only to orient the
 582 experimental context.

583 **Environmental transfer.** For each directed whole-screen evaluation, source controls comprised pooled
 584 cells from all other screens containing that reporter; source and destination groups were each determin-
 585 istically limited to 20,000 cells. For d phase features or reporter endpoints, non-targeting-control shift
 586 was

$$D_{r,s \rightarrow t}^{\text{NTC}} = \left\{ \frac{1}{d} \sum_{e=1}^d \left(\frac{\mu_{rt0e} - \mu_{rs0e}}{\sigma_{rs0e}} \right)^2 \right\}^{1/2}, \quad (7)$$

587 Coordinates with non-finite values or source-control standard deviation no greater than 10^{-8} were ex-
 588 cluded, where s denotes the pooled source and t the destination screen. Transfer loss was the median-
 589 across-methods held-out-gene score minus the corresponding strict destination-screen score. Associa-
 590 tions were quantified over 81 directions using Spearman correlation. For inference, phase shift, target
 591 shift and transfer loss were first summarized by their medians within each of the 34 reporters; two-sided
 592 P values were obtained by permuting the reporter-level shift summaries 3,000 times. Phase-control
 593 shift can be estimated before acquiring the destination targeted readout, whereas target-control shift is
 594 a retrospective explanatory quantity that requires that readout. A reporter was flagged as domain sen-
 595 sitive when its median cross-screen target stability was at or above the median among the 34 repeated
 596 reporters while its median strict-screen-minus-held-out-gene score was in the lowest quartile; the flag
 597 therefore marks an otherwise stable target with an unusually large model transfer penalty. Registered
 598 FeRhoNox control-cell images were deterministic phase and geometry medoids selected without refer-
 599 ence to transfer performance.

Cross-screen reporter stability. The context-stability analysis used all 81 frozen directed reporter-to-destination-screen evaluations for the 34 reporters observed in more than one screen. Recovery and magnitude-rank agreement were assessed separately. For each reporter and metric, all destination values at or above 0.70 were classified as stable above the gate, all values below 0.70 as stable below, and an observed range spanning 0.70 as crossing the gate. Supplementary Figure 4b shows the destination values, reporter median and observed range. This analysis did not recalculate the complete five-gate tier for each destination screen.

Conditional ambiguity. Conditional ambiguity was evaluated for FeRhoNox in two screens and pRb in one screen using 80,975 exact-linked, non-control cells. Reporter–screen–gene groups with fewer than six cells were excluded. The 172 phase variables were standardized within group, and each cell was assigned its nearest other cell by root-mean-square Euclidean distance; where multiple guides were available, the neighbour was required to carry a different guide. Target endpoints were standardized to the non-targeting-control median and median absolute deviation within reporter, screen and well, then clipped to $[-10, 10]$. Phenotype discrepancy was the root-mean-square endpoint difference. For each gene, the median discrepancy of the directed morphology-nearest pairs was divided by the median discrepancy of all eligible unordered pairs. Values near one indicate that close morphology matching leaves most targeted-state discrepancy intact. A 32-dimensional principal-component representation of raw phase thumbnails, fitted separately within each reporter and screen, provided a secondary representation-sensitivity analysis and was not treated as an image-learning upper bound.

Counterfactual response replacement and measurement-sufficiency classification

For each reporter, counterfactual response replacement substituted the observed knockout-response profile of every held-out gene with the element-wise unweighted median prediction across the ten aligned methods. Reporter labels from the training genes remained available during model fitting. The identical downstream workflow was then applied to observed and substituted held-out-gene profiles. The 50 genes with largest response magnitude (5% of the 1,000 knockout genes) were used for strong-hit recall and enrichment. Gene Ontology biological-process and cellular-component terms and Complex Portal terms containing 3–250 genes after mapping were tested by the hypergeometric distribution against the fixed source-atlas universe of 1,001 perturbation labels (1,000 knockout genes and NTC); NTC could not enter the 50-gene hit set. The ten terms with smallest unadjusted P values were retained, and observed–predicted agreement was their Jaccard overlap^{39–41}. These top-ten overlaps measure retention of ranked terms under the fixed workflow, not equivalence of biological discoveries. Annotation releases, gene identifiers and term universes were identical for observed and substituted profiles.

Let $P_{52}(z_r)$ denote the average-tie percentile rank of quantity z among the 52 reporters. The cohort-relative scientific-utility index was

$$U_r = \frac{1}{4} \left[P_{52}(H_r) + P_{52}(J_r^{\text{BP}}) + P_{52}(J_r^{\text{CC}}) + P_{52}(J_r^{\text{CP}}) \right], \quad (8)$$

where H is top-5% hit recall and the three J terms are top-ten overlaps for Gene Ontology biological process, Gene Ontology cellular component and Complex Portal. Magnitude ranking was excluded from U_r so that it could be evaluated as a separate predictor of downstream-analysis retention.

To relate fidelity to retained utility, we fitted Ridge regression using recoverability, magnitude-rank fidelity, magnitude-variance fidelity and within-screen reliability. Magnitude-variance fidelity transformed the variance ratio in Eq. 6 to $\exp\{-|\log Q_r^{\text{var}}|\}$, assigning the same value to reciprocal variance ratios. This transformation measures preservation of the spread of knockout-response magnitudes; it is not an intercept-and-slope calibration analysis. Evaluation used strict nested leave-one-reporter-out cross-validation. In each outer split, the four utility components for the 51 training reporters were converted to average-tie percentile ranks within that training set. Each held-out raw component was mapped to the corresponding training empirical distribution, with values outside the training range mapped to 0 or 1 and ties assigned the mean training percentile. The Ridge penalty was then selected in an inner leave-one-reporter-out loop that repeated the same training-only target construction. The held-out reporter’s downstream metrics were used only to construct its evaluation target against the

648 training reference distributions; they did not determine those distributions or model selection. Predic-
649 tive agreement was summarized by outer-loop Spearman correlation and R^2 ; univariate reporter-level
650 associations used the same strict outer targets.

651 The joint display in Fig. 6b used recoverability and the training-reference utility targets from the
652 reporter-heldout analysis for all 52 reporters. These utility values differ slightly from the descriptive
653 all-reporter percentile index in Fig. 6a,c. Its vertical line is the prespecified recoverability gate of 0.70;
654 its horizontal line is the cohort median utility and is descriptive only, not a decision threshold. The
655 recovery-utility association was evaluated using Spearman correlation across reporters. For rank statis-
656 tics and their bootstrap intervals in Fig. 6b and Supplementary Figure 4a, utility values were rounded
657 to 12 decimal places before assigning average ranks, preserving mathematical percentile ties against
658 floating-point perturbations; the underlying values and predictions were unchanged.

659 Candidate Ridge penalties in the nested analysis were $\alpha \in \{0.01, 0.1, 1, 10, 100\}$ and were selected by
660 mean inner-loop squared error.

661 **Measurement-sufficiency classification.** Rules were selected on this atlas, fixed before final tier assign-
662 ment and applied hierarchically. These tiers concern response-profile fidelity and perturbation prioriti-
663 zation and were generated directly from the following fidelity and reliability criteria; the cohort-relative
664 utility index in Eq. 8 was not used for tier assignment. A reporter was first classified as not identifiable
665 when within-screen reliability was unavailable or < 0.30 ; otherwise, missing required fidelity evidence
666 yielded unresolved. Among reporters with complete evidence, a quantitative proxy required held-out-
667 gene recoverability ≥ 0.70 , magnitude Spearman ≥ 0.70 , variance ratio from 0.50 to 1.50 and top-5%
668 recall ≥ 0.60 . A ranking proxy required magnitude Spearman ≥ 0.70 and top-5% recall ≥ 0.50 but did
669 not satisfy all quantitative criteria. Measurement required was assigned next when recoverability was
670 < 0.60 ; all remaining reporters were unresolved. This precedence allows a reproducible target that pre-
671 serves perturbation ranking to remain a ranking proxy even when its recoverability falls below 0.60. The
672 cut-offs are descriptive operating points for this atlas and intended uses, not cross-system standards;
673 their local sensitivity is reported in Supplementary Table 5. For a new application, tolerances should
674 be specified from the intended decision, error costs and relevant repeat measurements using develop-
675 ment data, then evaluated on held-out perturbations. Domain sensitivity was recorded as a separate
676 whole-screen-transfer flag and did not constitute a sixth tier.

677 To assess predictor dependence, we recalculated all four response criteria from each method's own
678 predictions, using the same checkpoint states as the response-replacement ensemble, and reapplied the
679 unchanged reliability values, thresholds and precedence. We distinguished quantitative-ranking trans-
680 sitions from changes across the proxy-use boundary, defined as quantitative or ranking proxy versus
681 measurement required or unresolved. The primary denominator was the 43 reporters with reliability
682 at least 0.30; the nine reliability-gated assignments were reported separately. The single-model bench-
683 mark states provided a second complete comparison, without retraining or selecting a best predictor
684 for each reporter. These comparisons describe the evaluated method set, not a sampling probability or
685 independent biological replication. Complete evidence and prediction-state sensitivity are provided in
686 Supplementary Data 1 and Supplementary Methods.

687 Independent applications

688 We surveyed public imaging and perturbation repositories and associated study deposits, recording 59
689 candidate resources by 12 August 2026. Each entry recorded six capabilities—a lower-cost broad phe-
690 notypic input, a targeted readout, paired observations, perturbations, within-screen negative controls
691 and repeated acquisition of the targeted readout—together with the first capability that determined el-
692 igibility and the strength of the supporting evidence. Forty-two resources could be classified from the
693 available evidence. Two were selected for analyses matched to the measurements and replication they
694 provided. Aggregate outcomes are in Supplementary Table 1; the complete 59-row registry with evi-
695 dence and exclusion reasons accompanies the source data.

696 The first selected resource, cpg0037-oasis, comprised 66 plates and 21,695 wells from a commercial
697 five-donor pool of primary human hepatocytes exposed to 1,085 non-control compounds⁴³. Pairing was
698 at the well rather than same-cell level. Inputs were 808 CellProfiler features attributable to the brightfield

channel; 120 brightfield-fluorescence cross-channel features, 4,512 fluorescence-attributable features and 200 channel-free features derived from segmentation were excluded. The retained features used fluorescence-derived object masks, making prediction conditional on the deposited segmentation. Primary targets were the deposited plate-normalized metabolic-activity and lactate-dehydrogenase-release endpoints. Compounds were randomly assigned to five outer folds with all wells and concentrations of a compound held out together. Outer-training-set median imputation and standardization preceded Ridge regression with fixed $\alpha = 1$. Recovery used the deposited plate-normalized endpoints. For response-fidelity analysis, observed and predicted well values were averaged by compound, plate and reporter, centred on the corresponding within-fold, plate-specific negative-control mean and then averaged across plates by compound. Reliability was Pearson correlation across compounds between two seeded, disjoint half-means of replicate non-control wells pooled across plates and concentrations, yielding a dose-averaged reference.

The second resource, cpg0021-periscope, paired DAPI, ConA, phalloidin and WGA Cell Painting measurements with anti-TOMM20 fluorescence in nine plates of A549-TetR-Cas9 cells¹¹. It was analysed at two resolutions: 1,080,000 single cells obtained by a deterministic cap of the first 120,000 streamed rows per plate, and all 704,771 deposited median-aggregated gene-sgRNA-plate profiles. Inputs comprised 2,508 measurements attributable exclusively to the four fluorescent predictor channels and targets comprised 823 anti-TOMM20-associated measurements; 60 predictor-target cross-channel and 375 channel-free features were excluded. Segmentation used phenotypic images that included the target channel, making this a cross-channel prediction analysis conditional on the deposited masks. Non-control genes were assigned to five outer folds; the next cyclic fold provided inner validation. Nontargeting controls remained in training. The Ridge penalty was selected by endpoint-standardized validation MSE and the model was refitted on the complete outer training set. Within each test fold, observed and predicted measurements were averaged by gene and plate and centred on the same-plate nontargeting-control mean. Recoverability was then calculated by dividing each endpoint's observed and predicted responses by its observed response standard deviation and pooling the gene-plate-endpoint coordinates for Pearson correlation. This endpoint-standardized pooled statistic differs from the equal-endpoint macro-average used for OPS (Eq. 2); endpoint inclusion and response-magnitude scaling are detailed in Supplementary Methods. Reliability split the sgRNAs for each eligible gene into two disjoint halves within plate, centred each half on plate-specific controls, standardized endpoints by the full-guide response spread and summarized the nine plate-level correlations by their median; genes with fewer than two sgRNAs and negligible-spread endpoints were excluded. This estimate concerns reproducibility of guide-defined gene responses and combines measurement, guide and biological variation.

The operational rules in the preceding subsection were applied to both resources without re-estimation. We assessed the dependence of these classifications on the numerical cut-offs using the same 36 threshold variations as in OPS, with predictions and metric aggregation unchanged (Supplementary Table 7 and Supplementary Data 2). Batch-specific analyses, reliability sensitivity, leave-one-dye-out analyses and within-screen input-permutation controls are described in Supplementary Methods.

Brightfield-guided metabolic-activity prioritization. The focused OASIS analysis used 768 features from the released brightfield-channel DINO representation and the deposited plate-normalized RealTime-Glo metabolic-activity readout. A fixed compound partition comprised 868 development compounds (13,157 wells) and 217 test compounds (3,300 wells); all measurements of test compounds were excluded from fitting across doses and production sources. The partition was inherited from a preceding representation exploration. The test compounds had not been evaluated in that brightfield-DINO exploration, although their target measurements were part of the public resource and its earlier CellProfiler analysis. The target, predictor settings, signed scoring rule and selection budgets were fixed before final fitting and held-out scoring. Brightfield inputs were centred on same-plate DMSO means, followed by training-only median imputation and standardization. Ridge regression used $\alpha = 100$ and all 768 features, without compound identity, concentration, fluorescence or cell-count inputs. Observed and predicted responses were each centred on their corresponding same-plate DMSO mean.

We defined loss as the negative control-relative response, so an increase in metabolic activity did not count as a loss. Within each compound and actual concentration, wells were averaged within produc-

tion source, then equally across available sources; these values were averaged equally across the eight assayed concentration positions. This score concerns the measured concentration panel, whose range can differ between compounds, rather than a common-range dose–response area. The primary selection budget was the top 5%, rounded up to 11 of 217 compounds; fixed secondary budgets were 10% and 20%. The 94-compound complete-source subset supplied dose-matched, cross-source prediction and measured-repeat comparisons, with five compounds selected at the primary budget. Compound bootstrap intervals used 2,000 resamples that preserved the dose and source pairing. They condition on the fitted predictor and observed sources. Predictor sensitivities, correspondence controls and complete source coverage are described in Supplementary Methods; predictions and selection records accompany Supplementary Data 3. The released representations came from brightfield channels of stained, fixed-cell acquisitions.

Label-efficiency and raw-phase representation analyses

Label-efficiency analysis. The low-label analysis (Supplementary Figure 2a) used a panel fixed before the learning-curve analysis—pS6, NPM1, SEC23A, LAMP1, 5xUPRE, TOMM20, FastAct, EEA1, WGA, Peroxi, Hoechst and G3BP1—and treated the remaining 40 reporters as a fully labelled donor atlas. The panel spans the ten biological-system annotations and includes reporters across the full-label recoverability range; it was not selected by low-label performance. Exact target-reporter observations were sampled independently within control-status-by-gene strata by Hamilton largest-remainder allocation, yielding exact budgets of 0.1%, 1% and 20% without consulting target values. The same fraction was applied to training and inner-validation cells; outer-test cells remained complete and sealed, and common sampling manifests were used across the five held-out-gene partitions. Ridge, gradient-boosted trees, CatBoost, the reporter-specific MLP, TabM, scButterfly and scPair used only the sparsely labelled target reporters. The shared residual MLP, MultiTab and MIDAS additionally used complete donor-atlas supervision. These two groups therefore represent target-only and atlas-assisted deployment settings rather than an equal-information ranking of algorithms. Equal-reporter partition means and standard deviations were calculated at each fraction. Matched 100% evaluations came from a separately completed full-label experiment on the same reporter panel and partitions. The connection from 20% to 100% was therefore treated as a cross-experiment calibration rather than a continuous within-run learning curve. Reporter identities and selected row indices for every partition and fraction are provided in the released sampling manifests.

Raw-phase representation analysis. The raw-phase representation analysis (Supplementary Figure 2b) used six reporters—EEA1, LAMP1, TOMM20, NPM1, pS6 and G3BP1—and five held-out-gene partitions. The reporters were fixed before representation comparison to span six biological systems and the full-label recoverability range, subject to availability of registered raw crops; they were not chosen by image-model performance. Registered 384×384 level-0 phase crops were encoded with a fixed pretrained DINOv2 ViT-S/14 vision transformer⁴⁷, a fixed pretrained Cytoland VSCyto2D encoder, a partially fine-tuned Cytoland encoder or the engineered 172-dimensional representation⁴⁸. Cytoland used the released VSCyto2D checkpoint. Valid pixels in each crop were centred by their median and scaled by their interquartile range; padded pixels were set to the valid-pixel median and no image augmentation was applied. For partial fine-tuning, the stem and first two encoder stages were fixed, the final two stages and a 256–128 reporter endpoint head were trained for at most 25 epochs using AdamW, batch size 48 and validation-based early stopping with patience 6. Backbone and head learning rates were 10^{-5} and 3×10^{-4} , respectively, with two warm-up epochs followed by cosine decay. Predictions were paired within each reporter and outer partition ($6 \times 5 = 30$ reporter–partition results per representation). Matrix entries show five-partition means and standard deviations within reporter. Comparative inference used the six reporters, with paired mean differences and the deposited reporter-hierarchical 95% bootstrap intervals. The complete reporter and crop manifests accompany the source data.

799 Statistical analysis and reproducibility

800 Pearson correlation measured linear association between observed and predicted phenotypes or re-
801 sponses; Spearman correlation measured rank agreement. Confidence intervals were obtained by re-
802 sampling the highest-level unit appropriate to the claim, generally reporter. Unless stated otherwise,
803 bootstrap intervals were percentile 95% intervals from 10,000 deterministic reporter resamples. Reporter-
804 to-screen analyses used reporter-level summaries or reporter-preserving resampling as specified above.
805 Multiplicity correction was applied only to the prespecified family of reporter-level method compar-
806 isons; permutation tests and bootstrap intervals for individual analyses were not multiplicity-adjusted.
807 Genes, endpoints, cells and partitions nested within reporter were not counted as independent reporter
808 replicates. Paired Wilcoxon tests and correlation permutation tests were two-sided; the joint-model R^2
809 permutation test used the upper tail of its null distribution. Exact sample sizes, exclusions, bootstrap
810 draws and permutation counts are given in the figure legends and panel-level source tables. Random
811 seeds and resampling draws are reproduced by the released workflows.

812 Software and hardware

813 Analyses used Python 3.10.20, NumPy 1.26.4, pandas 2.1.4, SciPy 1.11.4 and scikit-learn 1.7.2; neural
814 models used PyTorch 2.3.1, gradient-boosted trees used XGBoost 2.1.4 and CatBoost used version 1.2.10.
815 Computations used NVIDIA H100 GPUs (80 GB) and multi-core CPUs. Independent reporter- and fold-
816 level jobs were run in parallel, with up to four concurrent jobs per GPU in the same-cell analyses. Figures
817 were rendered with Matplotlib 3.7.5.

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820 Competing interests

821 The authors declare no competing interests.

822 Data availability

823 The optical perturbation atlas was generated by the OPS study³⁰. We used revision 6 of its public Zenodo
824 record (<https://doi.org/10.5281/zenodo.20495192>), released 1 June 2026 under CC BY 4.0⁴⁹. Full
825 imaging and molecular data remain available through the Biohub OPS Explorer (<https://biohub.ai/ops-explorer>)
826 under the source resource's terms. The two independent applications, cpg0037-oasis and
827 cpg0021-periscope, are available through the Cell Painting Gallery^{11,43,50}.

828 OASIS DINO profiles and metadata for metabolic-activity prioritization were obtained from the
829 source study's Axiom OASIS deposit (<https://doi.org/10.5281/zenodo.17067683>, Jessica Ewald, CC
830 BY 4.0). Derived tables retain source attribution and the deposited numerical values; predictions, aggrega-
831 tion and selection records are outputs of the present analysis.

832 Processed OPS inputs, pairing and split manifests, out-of-fold predictions and reporter-level results
833 are deposited in the MorphoSuff dataset repository ([https://huggingface.co/datasets/Amanda1998/](https://huggingface.co/datasets/Amanda1998/MorphoSuff)
834 [MorphoSuff](https://huggingface.co/datasets/Amanda1998/MorphoSuff)). Figure source data, registered example crops and Supplementary Data 1–3 are provided
835 in the GitHub repository listed below. These supplements contain predictor-specific decision evidence,
836 external threshold sensitivity and held-out metabolic-loss predictions and selections, respectively. Full
837 raw microscopy remains with the source resources; provenance and source-specific licences accompany
838 the released data.

839 Gene Ontology^{39,40}, Complex Portal⁴¹, CEGv2⁵¹ and MitoCarta 3.0⁵² were reused as public third-
840 party annotations. The releases, download dates and identifiers used in the analysis are recorded in the
841 source-data provenance.

842 Code availability

843 MorphoSuff code for data preparation, model fitting, evaluation and figure generation is available at
844 <https://github.com/limengran98/MorphoSuff> under the MIT License. The Python package provides
845 dataset adapters and shared response-fidelity, reliability and measurement-decision analyses for use
846 with other paired datasets or user-supplied predictions.

847 References

- 848 [1] Charlotte Bunne et al. How to build the virtual cell with artificial intelligence: Priorities and oppor-
849 tunities. *Cell*, 187(25):7045–7063, 2024. doi: 10.1016/j.cell.2024.11.015.
- 850 [2] Laralyn Przybyla and Luke A. Gilbert. A new era in functional genomics screens. *Nature Reviews*
851 *Genetics*, 23:89–103, 2022. doi: 10.1038/s41576-021-00409-w.
- 852 [3] Christoph Bock et al. High-content CRISPR screening. *Nature Reviews Methods Primers*, 2:8, 2022.
853 doi: 10.1038/s43586-021-00093-4.
- 854 [4] Michael Boutros, Florian Heigwer, and Christina Laufer. Microscopy-based high-content screening.
855 *Cell*, 163:1314–1325, 2015. doi: 10.1016/j.cell.2015.11.007.
- 856 [5] Mark-Anthony Bray et al. Cell Painting, a high-content image-based assay for morphological pro-
857 filing using multiplexed fluorescent dyes. *Nature Protocols*, 11:1757–1774, 2016. doi: 10.1038/nprot.
858 2016.105.
- 859 [6] Atray Dixit et al. Perturb-Seq: dissecting molecular circuits with scalable single-cell RNA profiling
860 of pooled genetic screens. *Cell*, 167:1853–1866.e17, 2016. doi: 10.1016/j.cell.2016.11.038.
- 861 [7] Joseph M. Replogle et al. Mapping information-rich genotype–phenotype landscapes with genome-
862 scale Perturb-seq. *Cell*, 185:2559–2575.e28, 2022. doi: 10.1016/j.cell.2022.05.013.
- 863 [8] Chris J. Frangieh et al. Multimodal pooled Perturb-CITE-seq screens in patient models de-
864 fine mechanisms of cancer immune evasion. *Nature Genetics*, 53:332–341, 2021. doi: 10.1038/
865 s41588-021-00779-1.
- 866 [9] YongKeun Park, Christian Depeursinge, and Gabriel Popescu. Quantitative phase imaging in
867 biomedicine. *Nature Photonics*, 12:578–589, 2018. doi: 10.1038/s41566-018-0253-x.
- 868 [10] David Feldman et al. Pooled genetic perturbation screens with image-based phenotypes. *Nature*
869 *Protocols*, 17:476–512, 2022. doi: 10.1038/s41596-021-00653-8.
- 870 [11] Meraj Ramezani et al. A genome-wide atlas of human cell morphology. *Nature Methods*, 22:621–633,
871 2025. doi: 10.1038/s41592-024-02537-7.
- 872 [12] Srinivasan Sivanandan et al. A pooled Cell Painting CRISPR screening platform enables de novo
873 inference of gene function by self-supervised deep learning. *Nature Communications*, 17:77, 2026.
874 doi: 10.1038/s41467-025-66778-6.
- 875 [13] Takamasa Kudo et al. Multiplexed, image-based pooled screens in primary cells and tissues with
876 PerturbView. *Nature Biotechnology*, 43:1091–1100, 2025. doi: 10.1038/s41587-024-02391-0.
- 877 [14] Juan C. Caicedo et al. Data-analysis strategies for image-based cell profiling. *Nature Methods*, 14:
878 849–863, 2017. doi: 10.1038/nmeth.4397.
- 879 [15] Xinyang Li, Yuanlong Zhang, Jiamin Wu, and Qionghai Dai. Challenges and opportunities in bioim-
880 age analysis. *Nature Methods*, 20:958–961, 2023. doi: 10.1038/s41592-023-01900-4.
- 881 [16] Srinivas Niranj Chandrasekaran, Hugo Ceulemans, Justin D. Boyd, and Anne E. Carpenter. Image-
882 based profiling for drug discovery: due for a machine-learning upgrade? *Nature Reviews Drug*
883 *Discovery*, 20:145–159, 2021. doi: 10.1038/s41573-020-00117-w.
- 884 [17] Gregory P. Way et al. Predicting cell health phenotypes using image-based morphology profiling.
885 *Molecular Biology of the Cell*, 32:995–1005, 2021. doi: 10.1091/mbc.E20-12-0784.
- 886 [18] Eric M. Christiansen et al. In silico labeling: predicting fluorescent labels in unlabeled images. *Cell*,
887 173:792–803.e19, 2018. doi: 10.1016/j.cell.2018.03.040.

- [19] Chawin Ounkomol, Sharmishta Seshamani, Mary M. Maleckar, Forrest Collman, and Gregory R. Johnson. Label-free prediction of three-dimensional fluorescence images from transmitted-light microscopy. *Nature Methods*, 15:917–920, 2018. doi: 10.1038/s41592-018-0111-2.
- [20] Gyuhyun Lee, Jeong-Woo Oh, Nam-Gu Her, and Won-Ki Jeong. DeepHCS++: bright-field to fluorescence microscopy image conversion using multi-task learning with adversarial losses for label-free high-content screening. *Medical Image Analysis*, 70:101995, 2021. doi: 10.1016/j.media.2021.101995.
- [21] Jan Oscar Cross-Zamirski et al. Label-free prediction of Cell Painting from brightfield images. *Scientific Reports*, 12:10001, 2022. doi: 10.1038/s41598-022-12914-x.
- [22] Rahil Mehrizi et al. Multi-omics prediction from high-content cellular imaging with deep learning. *arXiv preprint arXiv:2306.09391*, 2023.
- [23] Karren Dai Yang et al. Multi-domain translation between single-cell imaging and sequencing data using autoencoders. *Nature Communications*, 12:31, 2021. doi: 10.1038/s41467-020-20249-2.
- [24] Justin Lakkis et al. A multi-use deep learning method for CITE-seq and single-cell RNA-seq data integration with cell surface protein prediction and imputation. *Nature Machine Intelligence*, 4:940–952, 2022. doi: 10.1038/s42256-022-00545-w.
- [25] Marzieh Haghighi, Juan C. Caicedo, Beth A. Cimini, Anne E. Carpenter, and Shantanu Singh. High-dimensional gene expression and morphology profiles of cells across 28,000 genetic and chemical perturbations. *Nature Methods*, 19:1550–1557, 2022. doi: 10.1038/s41592-022-01667-0.
- [26] Safiye Celik et al. Building, benchmarking, and exploring perturbative maps of transcriptional and morphological data. *PLOS Computational Biology*, 20:e1012463, 2024. doi: 10.1371/journal.pcbi.1012463.
- [27] Jianxu Chen, Matheus P. Viana, and Susanne M. Rafelski. When seeing is not believing: application-appropriate validation matters for quantitative bioimage analysis. *Nature Methods*, 20:968–970, 2023. doi: 10.1038/s41592-023-01881-4.
- [28] Srijit Seal et al. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nature Communications*, 17:2436, 2026. doi: 10.1038/s41467-026-68725-5.
- [29] Nitsan Elmalam and Assaf Zaritsky. Cell context-dependent in silico organelle localization in label-free microscopy images. *Nature Methods*, 23:405–416, 2026. doi: 10.1038/s41592-025-02960-4.
- [30] Chad Liu et al. A multimodal perturbation atlas defines the phenotypic resolution of cellular morphology. *bioRxiv*, 2026. doi: 10.64898/2026.06.01.728087.
- [31] Srinivas Niranj Chandrasekaran et al. JUMP Cell Painting dataset: morphological impact of 136,000 chemical and genetic perturbations. *bioRxiv*, 2023. doi: 10.1101/2023.03.23.534023.
- [32] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, pages 6638–6648, 2018.
- [33] Yury Gorishniy, Akim Kotelnikov, and Artem Babenko. TabM: Advancing tabular deep learning with parameter-efficient ensembling. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [34] Yichuan Cao et al. scButterfly: a versatile single-cell cross-modality translation method via dual-aligned variational autoencoders. *Nature Communications*, 15:2973, 2024. doi: 10.1038/s41467-024-47418-x.

- [35] Zhen He et al. Mosaic integration and knowledge transfer of single-cell multimodal data with MIDAS. *Nature Biotechnology*, 42:1594–1605, 2024. doi: 10.1038/s41587-023-02040-y.
- [36] Hongru Hu and Gerald Quon. scPair: Boosting single cell multimodal analysis by leveraging implicit feature selection and single cell atlases. *Nature Communications*, 15:9932, 2024. doi: 10.1038/s41467-024-53971-2.
- [37] Shaliu Fu et al. Benchmarking single-cell multi-modal data integrations. *Nature Methods*, 22:2437–2448, 2025. doi: 10.1038/s41592-025-02737-9.
- [38] Yan Wu et al. PerturBench: Benchmarking machine learning models for cellular perturbation analysis. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-3225.
- [39] Michael Ashburner et al. Gene ontology: tool for the unification of biology. *Nature Genetics*, 25: 25–29, 2000. doi: 10.1038/75556.
- [40] Gene Ontology Consortium. The Gene Ontology knowledgebase in 2023. *Genetics*, 224:iyad031, 2023. doi: 10.1093/genetics/iyad031.
- [41] Birgit H M Meldal et al. Complex Portal 2022: new curation frontiers. *Nucleic Acids Research*, 50: D578–D586, 2022. doi: 10.1093/nar/gkab991.
- [42] Stefan Peidli et al. scPerturb: harmonized single-cell perturbation data. *Nature Methods*, 21:531–540, 2024. doi: 10.1038/s41592-023-02144-y.
- [43] Jessica D. Ewald et al. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Systems*, 17(5):101566, 2026. doi: 10.1016/j.cels.2026.101566.
- [44] Michael Doron et al. Unbiased single-cell morphology with self-supervised vision transformers. *bioRxiv*, 2023. doi: 10.1101/2023.06.16.545359.
- [45] Oren Kraus et al. Masked autoencoders for microscopy are scalable learners of cellular biology. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11757–11768, 2024. doi: 10.1109/cvpr52733.2024.01117.
- [46] Kian Kenyon-Dean et al. ViTally consistent: Scaling biological representation learning for cell microscopy. *Proceedings of Machine Learning Research*, 267:29735–29752, 2025.
- [47] Maxime Oquab et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.
- [48] Ziwen Liu et al. Robust virtual staining of landmark organelles with Cytoland. *Nature Machine Intelligence*, 7:901–915, 2025. doi: 10.1038/s42256-025-01046-2.
- [49] Chad Liu et al. A multimodal perturbation atlas defines the phenotypic resolution of cellular morphology. Zenodo, 2026.
- [50] Erin Weisbart et al. Cell Painting Gallery: an open resource for image-based profiling. *Nature Methods*, 21(10):1775–1777, 2024. doi: 10.1038/s41592-024-02399-z.
- [51] Traver Hart et al. Evaluation and design of genome-wide CRISPR/SpCas9 knockout screens. *G3 Genes | Genomes | Genetics*, 7(8):2719–2727, 2017. doi: 10.1534/g3.117.041277.
- [52] Sneha Rath et al. MitoCarta3.0: an updated mitochondrial proteome now with sub-organelle localization and pathway annotations. *Nucleic Acids Research*, 49(D1):D1541–D1547, 2021. doi: 10.1093/nar/gkaa1011.